

changes, is not typically beneficial and surgical management with Roux-en-y gastrojejunostomy is often indicated



RESULTS

Truncal vagotomy with antrectomy is highly effective in reducing both basal and stimulated acid secretion in the stomach, as well as highly effective in reducing ulcer recurrence. Recurrence rates of 1% to 2% have been reported after truncal vagotomy and antrectomy, even prior to the use of treatment for *H. pylori*.



CONCLUSIONS

Truncal vagotomy and antrectomy is an effective means of treating refractory symptoms of peptic ulcer disease, particularly with pyloric channel ulcers, gastric outlet obstruction, or in the acute setting in a patient with ulcer disease while on antisecretory medications. In the setting of highly effective antisecretory therapy and the recognition and treatment of *H. pylori* colonization, surgical therapy is rarely required as an acid-reducing intervention. However, these procedures remain an important tool for the gastrointestinal surgeon treating refractory ulcer disease or faced with emergency complications of peptic ulcer disease such as perforation, bleeding, or obstruction.

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4 Laparoscopic Truncal Vagotomy with Antrectomy and Billroth I Reconstruction

Aurora D. Pryor



INDICATIONS/CONTRAINDICATIONS

Peptic Ulcer Disease

Surgical management of peptic ulcer disease has metamorphosed with the advent of proton pump inhibitor (PPI) therapy and elucidation of the role of *Helicobacter pylori* in gastric pathology. While gastric resections for ulcer disease were formerly a mainstay in the practice of general surgery, such procedures are now rare. Most patients are treated with 6 weeks of PPI therapy and eradication of *H. pylori*. This combination of therapies is effective in treating the majority of ulcers. However, indications do remain for surgical therapy for ulcer disease. The major indication is failure of medical management presenting as either intractable ulceration after a 3-month medical trial or issues such as perforation or bleeding while on therapy. Ulcers can also present acutely as a complication of nonsteroidal anti-inflammatory drug use. Long-term complications of ulcers, such as gastric outlet obstruction or malignancy, remain reasons for surgery as well. An additional relative indication for surgery is a patient with complicated ulcer disease and poor access to care or unlikely follow-up.

The choice of surgical procedure is based on the indication. Simple Graham (omental) patch closure can be performed for pyloric channel or duodenal perforation. Vagotomy and pyloroplasty offers a definitive approach to a bleeding pyloric channel or duodenal ulcer. Wedge gastrectomy results in excision of perforated gastric ulcers, while providing adequate tissue to evaluate for underlying malignancy. Antrectomy, usually in conjunction with vagotomy, is most commonly used for patients with gastric outlet obstruction from chronic ulcer disease. Vagotomy and antrectomy can also be used for definitive management of a perforated pyloric channel ulcer in a stable patient with minimal soilage.

Vagotomy is performed to minimize acid production in patients undergoing surgery for peptic ulcer disease. It works by blocking acetylcholine-mediated acid release as

well as preventing antral interneuron release of gastrin-releasing peptide which in turn stimulates antral G-cell gastric release. However, histamine and nonvag-mediated gastrin release still occur following vagotomy. Truncal vagotomy gained popularity in the 1940s in the management of gastric acid production to limit ulcer formation. It was performed in conjunction with a drainage procedure to minimize complications of delayed gastric emptying that were witnessed in up to one-third of patients following vagotomy alone. Selective and highly selective vagotomy came into favor to maximize the benefits of diminished acid production without causing gastric dysmotility. With the success of PPI therapy, even these selective procedures have fallen out of favor. Most patients requiring surgery for ulcer disease today require either antrectomy for obstruction or duodenal access and therefore pyloroplasty closure. In these patients, concomitant truncal vagotomy can obviate the need for and expense of postoperative PPI with minimal added risk.

Gastrointestinal Stromal Tumors

Gastrointestinal stromal tumors (GISTs) are mesenchymal neoplasms originating from the cells of Cajal. Sixty percent of these tumors are found in the stomach. The diagnosis is suspected with any submucosal mass and confirmed in most cases with reactivity to KIT (CD117) and/or DOG1. Many GISTs are incidentally found, but some present as a source of bleeding due to rupture. The malignant potential increases with both size and mitotic index, with tumors <2 cm in diameter and with ≤ 5 mitoses per 50 HPF considered very low risk. Surgery is the mainstay of treatment for localized GISTs amenable to resection.

Surgical resection for GISTs involves complete margin-negative resection with an intact pseudocapsule. Lymphadenectomy is not required. Although most of these lesions may be treated with wedge resection, occasionally antrectomy is required.

Pancreatic rests and adenomas may present preoperatively as suspected GISTs. Although they have low malignant potential, they are usually managed as if GISTs unless a benign diagnosis can be confirmed.

Carcinoids

Gastric neuroendocrine tumors are an unusual cause of gastric pathology. These lesions can be benign carcinoid tumors or neuroendocrine carcinomas. This determination can usually be made on the basis of hematoxylin- and eosin-stained biopsies. Some carcinoid lesions may be associated with atrophic gastritis, pernicious anemia, or elevated gastrin levels. Most small lesions confirmed to be benign can be removed with endoscopic or laparoscopic wedge techniques. Some advocate surveillance for benign lesions 10 to 20 mm or smaller. Many suggest that multifocal carcinoids should be resected with antrectomy. Neuroendocrine carcinomas should be managed with more radical resection, often total gastrectomy with lymphadenectomy.

Adenocarcinoma

Gastric adenocarcinoma is rarely confined to just the distal stomach in Europe or the Americas. The surgical approach is geared at obtaining an R0 resection, with negative margins and no evidence of residual disease whenever possible. In the United States and Europe, D1 lymphadenectomy involving just the perigastric nodes is standard. More extensive D2 lymphadenectomy adding nodal tissue surrounding the left gastric, splenic, and hepatic arteries as well as the celiac axis, although beneficial in Japan, has been associated with increased morbidity and mortality in Western trials. Removal of at least 15 lymph nodes is important for staging information.

Outcomes after laparoscopic versus open gastric resection are oncologically equivalent based on several studies in the setting of experienced laparoscopic surgeons and low-risk patients (ASA < 3). There are benefits with length of stay, resumption of oral intake, blood loss, and analgesia use as well.

On the basis of results of a landmark 2006 study by Cunningham and colleagues, patients with gastric or gastroesophageal adenocarcinoma should receive preoperative chemotherapy to improve progression-free and overall survival.

For patients with advanced disease, palliative resection, even involving total gastrectomy, has improved outcomes compared with surgical bypass. This includes both quality of life and possibly survival benefits.

PREOPERATIVE PLANNING

Chronic ulcer disease patients require a trial of medical therapy prior to surgery. Most patients are expected to complete two 6-week courses of PPI therapy as well as eradication of *H. pylori* with proven failure. No specific bowel prep is required.

For patients with acute complications of ulcer disease, such as bleeding or perforation, preoperative resuscitation is important. These patients should have a urinary catheter placed. Blood or fluids should be given as necessary to assure adequate perfusion. In the setting of perforation, broad-spectrum antibiotics are also advised. A nasogastric tube may be placed to facilitate gastric decompression, but it should be removed before stapling.

SURGERY

Positioning

The setup for laparoscopic antrectomy in our practice has the patient placed supine with the arms extended out at the sides (Fig. 4.1). The surgeon stands at the patient's right, with the assistant on the left. Monitors are placed over the patient's shoulders.

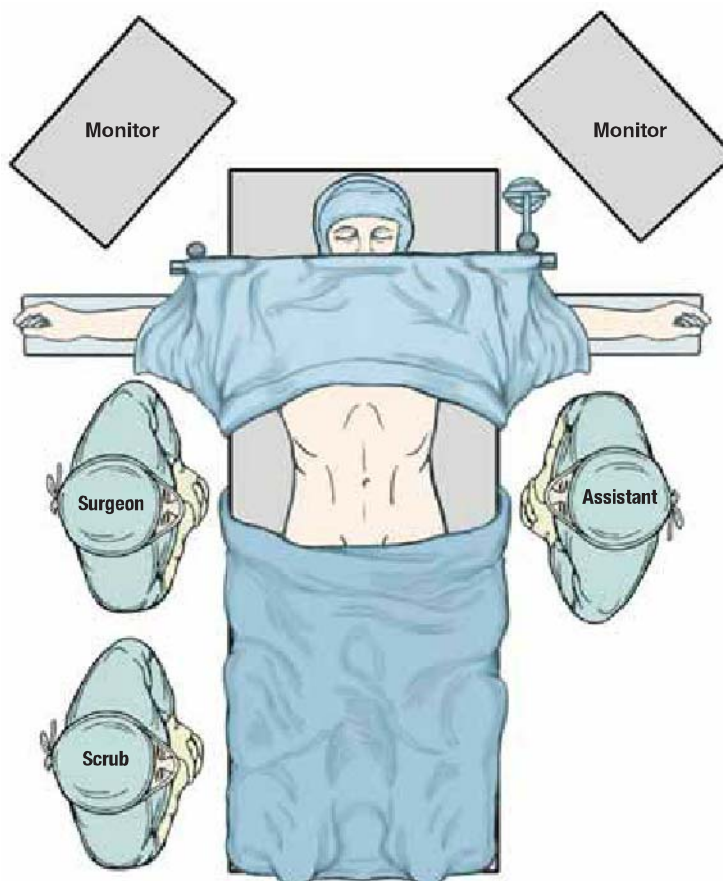
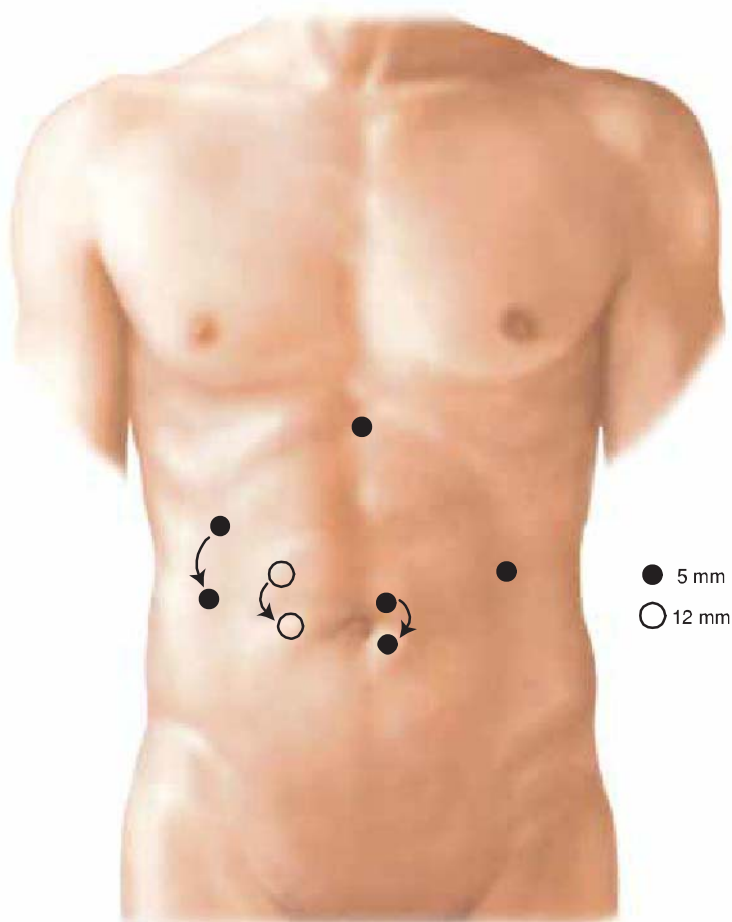


Figure 4.1 OR setup for laparoscopic gastrectomy. The patient is placed supine on the table with arms extended. The surgeon stands to the patient's right and the assistant is on the left.

Figure 4.2 Port placement for laparoscopic gastrectomy. The ports are placed higher for proximal pathology or lower (arrows) for more distal lesions.



An endoscope should be readily available to help with identification of pathology or to test for leak at the completion of the procedure. Ports are placed to allow good access to the hiatus and distal stomach (Fig. 4.2). The procedure is performed in reverse Trendelenburg position. A liver retractor is helpful for proximal gastric and hiatal exposure. This is placed through a subxiphoid port.



SURGICAL TECHNIQUE

Distal Gastrectomy

Distal gastrectomy may be performed for ulcer disease, benign lesions as well as cancer. If the resection is being performed for adenocarcinoma, the resection should include the greater omentum, duodenal bulb, and surrounding nodal tissue. The technical ease of antrectomy is greatly improved with modern vessel sealing technology. These devices can divide vessels up to 7 mm in diameter without requiring skeletonization or clips.

The variability of distal gastrectomy is greatly dependent on the planned reconstruction. The procedure can be completed with a gastroduodenostomy (Billroth I anastomosis), loop gastrojejunostomy (Billroth II anastomosis), or with Roux-en-Y reconstruction (Fig. 4.3). Billroth I reconstruction is the most anatomic and is preferred in our practice when technically possible. It also minimizes postoperative complications such as duodenal stump leak, afferent loop syndrome, and marginal ulceration. This approach will be the focus of this chapter. The anastomosis may be placed in a

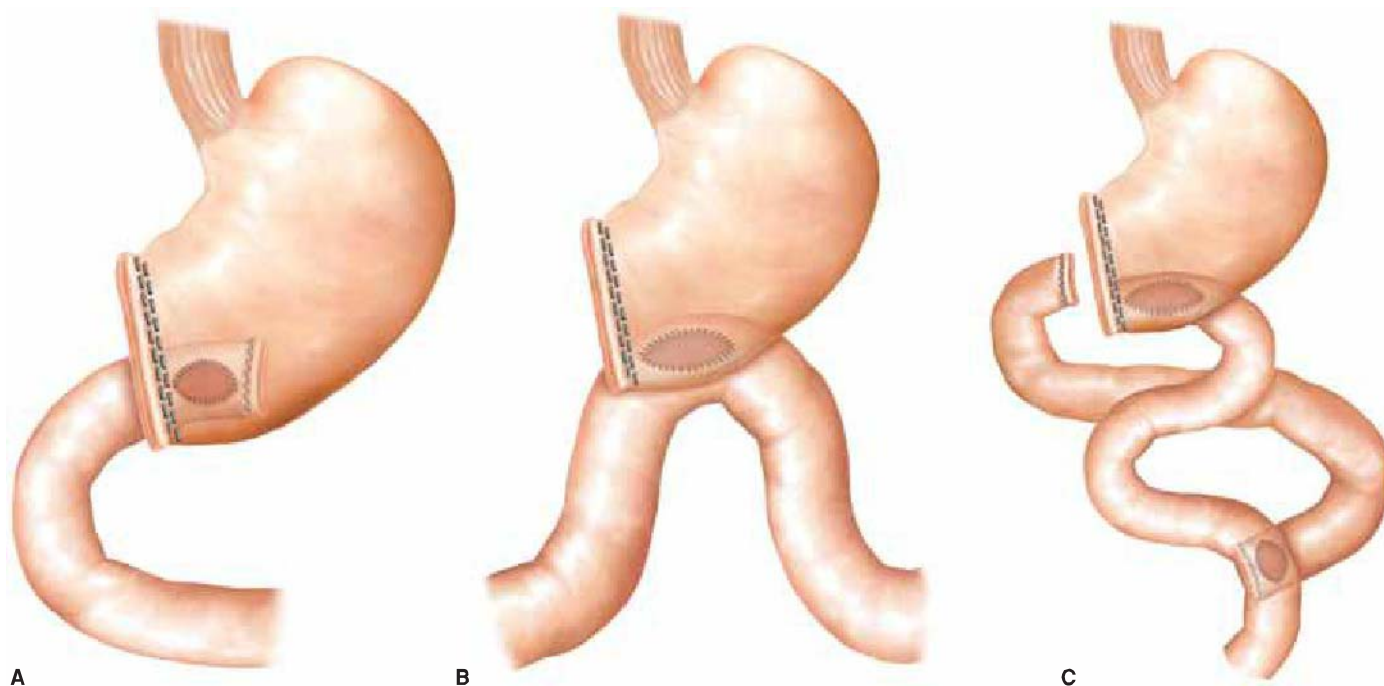


Figure 4.3 Alternatives for reconstruction after distal gastrectomy. (A) Billroth I gastroduodenostomy. (B) Billroth II loop gastrojejunostomy, or (C) Roux-en-Y reconstruction.

variety of locations, but the distal posterior gastric wall is usually preferred to facilitate gastric emptying.

- The procedure should begin with duodenal mobilization (Kocher maneuver, Fig. 4.4). The more extensive this mobilization, the easier the reach for the eventual anastomosis.
- Omental mobilization is next if the procedure is performed for cancer. This begins with separating the omentum off of the transverse colon and carrying this mobilization up to the stomach.
- The transection line on the stomach is then identified. Antrectomy is usually performed proximal to the crow's foot, such as insertion of vessels onto the lesser curvature and at a point corresponding to the transition between the right and left gastroepiploic vessels on the greater curvature. The line between these points will include the entire antrum.
- The gastroepiploic vessels are divided along the greater curvature at the transection plane and the lesser sac window is confirmed (Fig. 4.5). After removing any nasogastric tube, the stomach is divided with staples (Fig. 4.6). In our practice we prefer a bioabsorbable buttress material and 4.8-mm staple leg lengths for this division (Duet, Covidien, North Haven, CT). Other staple options can be used per surgeon preference. The transection is continued toward the lesser curvature.

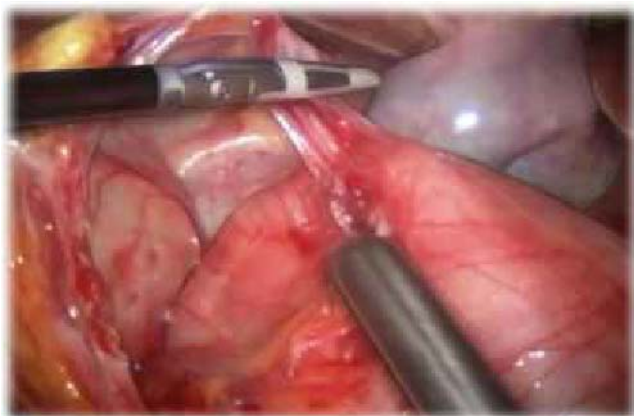


Figure 4.4 Laparoscopic Kocher maneuver.

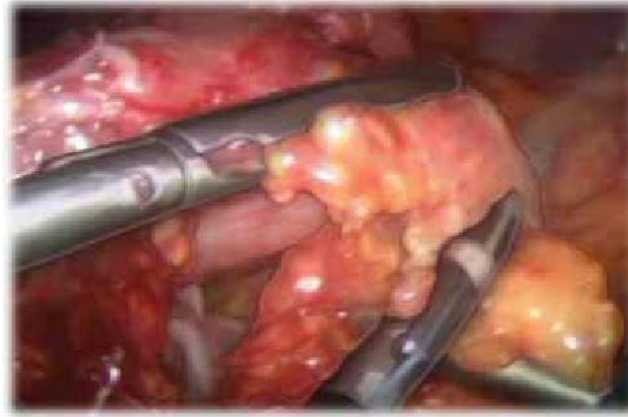


Figure 4.5 Mobilizing the antrum for resection.

- After fully dividing the stomach, the tissue along the lesser curvature is separated with a bipolar sealing device, care being taken to keep the nodal tissue with the specimen when operating for cancer.
- The distal stomach then functions as a handle to assist with pyloric and duodenal mobilization. It is reflected to the patient's right to facilitate exposure. Surrounding nodal tissue is kept with the specimen.
- The right gastric and right gastroepiploic vessels may be divided with vessel sealing. Alternatively, staples may be used.
- The duodenum is then divided with 2.5-mm staples (Fig. 4.7). Staple line reinforcement or oversewing of this staple line should be considered if the duodenum will not be used for reconstruction.
- The specimen is then placed in a retrieval bag for removal at the completion of the procedure.

Vagotomy

Exposure for vagotomy mimics hiatal exposure for other procedures involving the gastroesophageal junction.

- The lesser omentum is incised with an electrosurgical instrument and the hiatus is exposed.
- This dissection is carried anterior to the esophagus to mobilize the gastroesophageal junction.
- The anterior vagus nerve is identified overlying the esophagus, toward the patient's left. It can usually be palpated and separated easily from the esophagus with careful dissection.
- Traditionally, clips have been placed proximally and distally on the vagus nerve to mark the vagotomy. This may be performed at the surgeon's discretion.



Figure 4.6 Dividing the stomach.

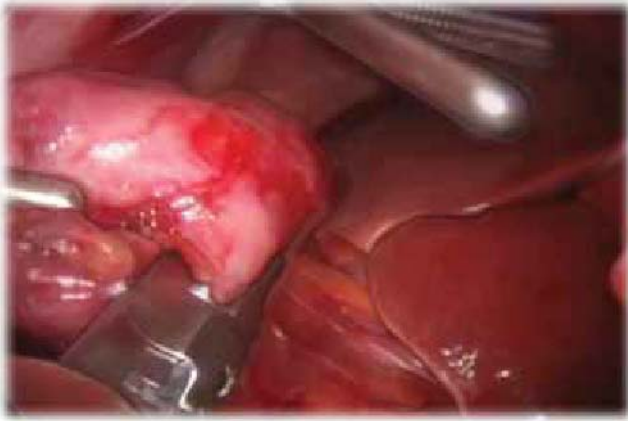


Figure 4.7 Dividing the duodenum with staples.

- The nerve is divided with scissors, bipolar sealing, or ultrasonic dissection (Fig. 4.8). It is then removed from the operative field and labeled.
- A plane is then created posterior to the esophagus, adjacent to the right crus.
- The esophagus is retracted anteriorly and to the patient's left to facilitate adequate visualization.
- The posterior vagus nerve can usually be identified running between the esophagus and the right crus. It can be dissected off the esophagus with gentle dissection.
- The nerve should be divided as high in the chest as possible to assure that the criminal nerve of Grassi is divided. Clips are again used at the surgeon's discretion (Fig. 4.9).
- The nerve is then removed from the operative field and labeled.
- The specimens should be sent to pathology for confirmation of peripheral nerve on frozen section. Anterior and posterior nerve candidates should be carefully labeled. While waiting for results of the frozen section, reconstruction may be performed.

Billroth I Reconstruction

If the scarring near the duodenum is minimal and the resection confined to the antrum, Billroth I reconstruction is reasonably straightforward. Gastroduodenostomy can be completed with linear staples, circular staples, or a hand-sewn anastomosis. In our practice a linear stapled anastomosis is preferred.

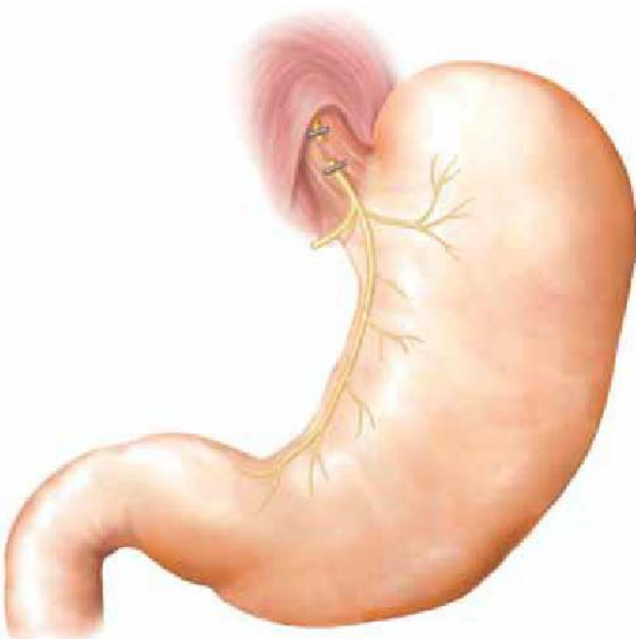


Figure 4.8 Completing the anterior vagotomy.

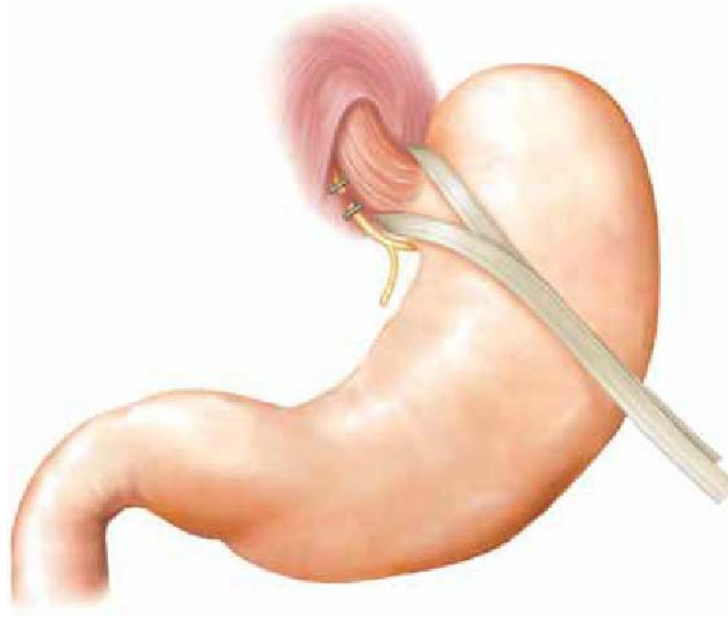


Figure 4.9 Posterior vagotomy. It is essential to get high on the esophagus (at least 5 cm) to include the criminal nerve of Grassi in the vagotomy. The exposure is similar to that used for a hiatal hernia repair.

- To complete this anastomosis, a gastrotomy is placed on the posterior stomach near the inferior edge of the staple line (Fig. 4.10).
- A duodenotomy is created on the anterior surface of the duodenum with adequate distance to the duodenal staple line to accommodate a linear stapled anastomosis (at least 3 cm) (Fig. 4.11).
- The 3.5-mm leg-length endoscopic linear stapler is then placed into the duodenotomy and brought toward the stomach, advancing the anterior staple anvil leg into the gastrotomy.
- The stapler is then fired (Fig. 4.12).
- A suture is placed on the posterior gastric wall and duodenum at the end of the staple line to diminish any tension.
- The remaining enterotomy is closed with sutures in a running fashion. To facilitate this, we place a post suture at the distal end of the anastomosis, incorporating both full thickness stomach and duodenum (Fig. 4.13).
- We then start with a longer (25 cm) suture at the other end of the anastomosis and run with a single suture to the original post (Fig. 4.14). This completes a single-layer anastomosis (Fig. 4.15).
- An endoscope is placed transorally to evaluate the anastomosis for bleeding and leak. The anastomosis is submerged while the endoscope is used for insufflation.
- A second layer of sutures is placed only if the leak test is positive. Interrupted sutures may be placed to contain any area of air leak.

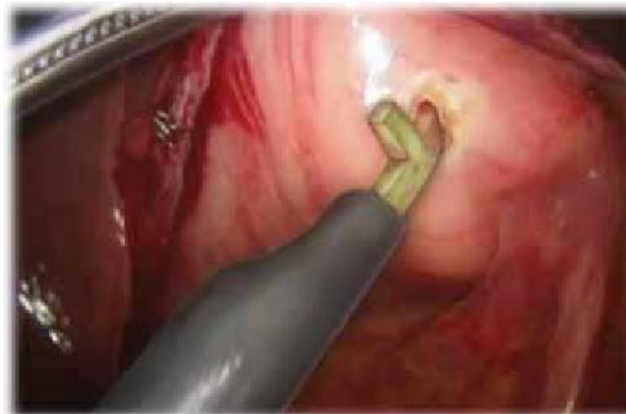


Figure 4.10 Creating a gastrotomy on the posterior stomach near the inferior border of the staple line.

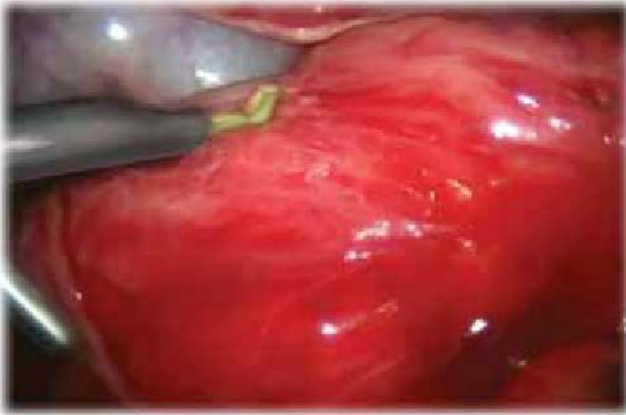


Figure 4.11 Creating a duodenotomy with adequate space to allow for passage of the stapler proximally.

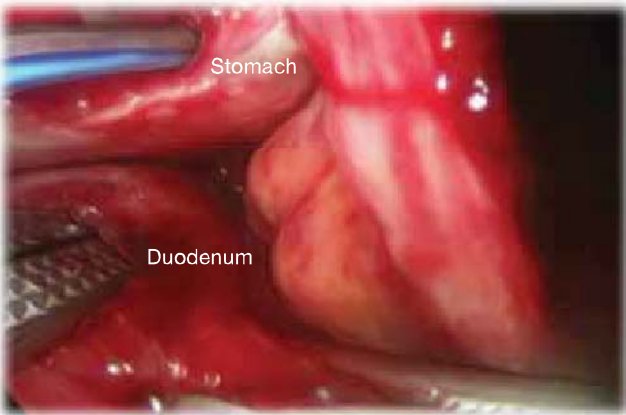


Figure 4.12 Creating the gastrodudenostomy with a linear stapler.

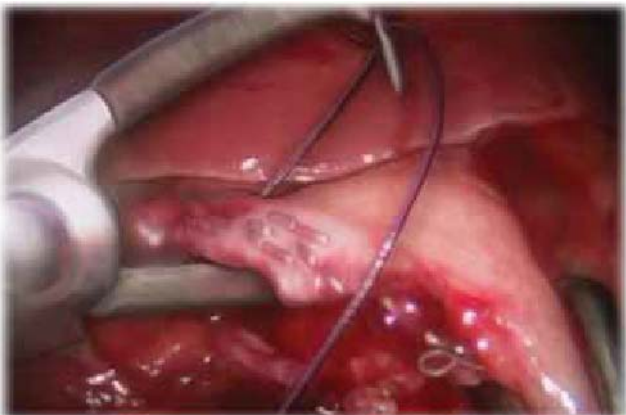


Figure 4.13 Placing a post stitch across the posterior portion of the gastrodudenostomy staple line.

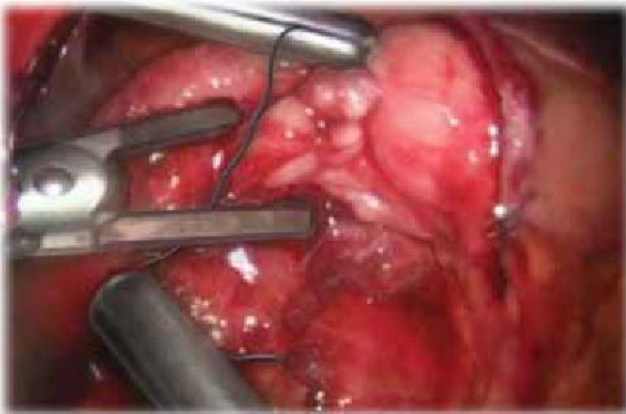


Figure 4.14 Suturing the common enterotomy on a gastrodudenostomy.

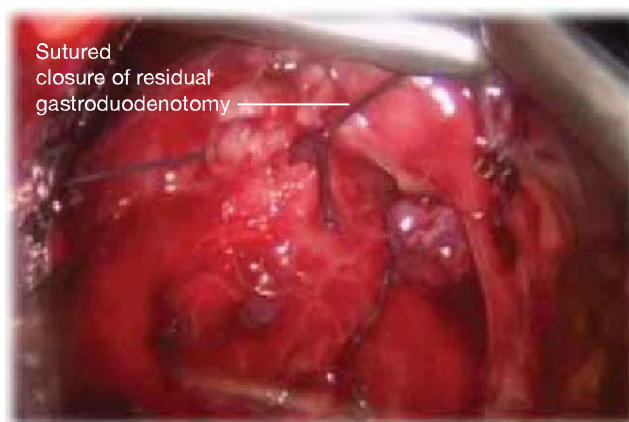


Figure 4.15 Completing the gastroduodenostomy.

Use of Endoscopic Adjuncts

Some surgeons have found that endoscopic visualization is helpful during transgastric resections. We have found it difficult to keep the field of view level and appropriately oriented when using the endoscope exclusively. We feel the endoscope is helpful for confirming pathology and assuring adequacy of final closure with a negative leak test.



POSTOPERATIVE MANAGEMENT

For most gastric resections, we do not place drains or nasogastric tubes. Exceptions to this are for high-risk patients (malnutrition, chemotherapy, chronic gastric outlet obstruction) or high-risk anastomoses (poor tissue, tension). In straightforward procedures, if the patients look and feel good on the morning of postoperative day 1, a liquid diet is implemented. Diet is advanced to semisolid foods at 2 weeks. If there are clinical signs of difficulty such as fever or tachycardia, or other reasons for clinical suspicion, a water-soluble contrast upper GI is obtained. Patients are discharged as early as postoperative day 1.



COMPLICATIONS

Procedure-related perioperative complications may include bleeding, leak, or esophageal injury, although these are not commonly reported. Complications of open gastrectomy are seen in the laparoscopic population as well. These include alkaline reflux gastritis in 1% to 2%, gastroparesis in 2% (particularly if resection is accompanied by vagotomy), dumping syndrome, and postvagotomy diarrhea in 2% to 4%.



RESULTS

Although few large laparoscopic series exist, most patients do well after laparoscopic antrectomy. Truncal vagotomy and antrectomy is curative for 99% of ulcer disease patients. Outcome for other pathology depends on the adequacy of the resection for cure. Recurrence is unlikely for low-grade GISTs and benign carcinoids. For results in cancer, see Table 4.1.

TABLE 4.1 Laparoscopic-Assisted Distal Gastrectomy (LADG) for Early Gastric Cancer

Year	N	Procedure	Extent of Resection	Complications	Follow-up (mo)	Recurrence
2002	14	LADG	D1	Pancreatic injury, 1 delayed gastric emptying, 1	24 ± 9.6	None
2001	7	LADG	D1	None	17	None
2003	76	LADG	D1	Anastomotic leakage, 3 Abscess, 5 Postoperative bleeding, 1	29	None
2002	116	LADG	D1	Pneumonia, 1 Anastomotic leakage, 1 Pancreatic injury, 1 Anastomotic stenosis, 1	45	None
2003	43	LADG	D1	Anastomotic leakage, 6 Wound infection, 2 Intraoperative hemorrhage, 2 Anastomotic ulcer, 1	37 ± 7	One systemic recurrence

From: Fischer JF and Bland KI, editors. *Mastery of Surgery*. 5th ed. Volume I. Philadelphia, PA: Lippincott, Williams and Wilkins.



CONCLUSIONS

- Antrectomy is appropriate for a variety of gastric pathologies.
- If antrectomy is performed for ulcer disease, it is due to acute complications or medical failure.
- Truncal vagotomy is added when performing antrectomy for ulcer disease.
- Laparoscopic antrectomy can be facilitated with bipolar sealing and linear stapling.
- Reconstruction after antrectomy can be with gastroduodenostomy, loop gastrojejunostomy, or Roux-en-Y gastrojejunostomy.
- For patients without duodenal scarring, gastroduodenostomy is straightforward and the most anatomic anastomosis.
- Gastroduodenostomy can be performed with linear staplers, with suture closure of the remaining gastroduodenostomy.
- Outcomes following laparoscopic antrectomy +/- truncal vagotomy are excellent, with minimal recurrence for benign pathologies.

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5 Truncal Vagotomy with Antrectomy and Billroth II Reconstruction

Michael W. Mulholland



INDICATIONS

Truncal vagotomy and antrectomy is usually performed in the treatment of complications of peptic ulcer disease. Indications include perforation, hemorrhage, and obstruction. The elucidation of the pathogenic role of *Helicobacter pylori* in the development of peptic ulceration has decreased the need for operative treatment of peptic ulceration. In addition, the wide availability of proton pump inhibitors and histamine-2 receptor blockers has further diminished the need for truncal vagotomy, an operation designed to permanently suppress gastric acid production.

When truncal vagotomy and antrectomy is used, the first goal is safety from immediate surgical complications and avoidance of long-term undesirable side effects. The second goal is repair of associated anatomic defects, such as pyloric obstruction.



PREOPERATIVE PLANNING

Preparation for elective antrectomy should seek to optimize physiologic status. Electrolyte imbalances, common with gastric outlet obstruction, should be corrected. Retained food and particulate matter may also accumulate within an obstructed stomach. These materials can cause bacterial overgrowth and should be evacuated preoperatively. Scrupulous attention to general nutrition, anemia, and associated comorbid conditions is also required preoperatively.

Vagal Anatomy

The vagal nerves form a plexus which surrounds the intrathoracic esophagus. The nerve trunks coalesce within the thorax before entering the abdomen through the esophageal

hiatus. The anterior (left) vagus nerve is positioned in immediate contact with anterior wall of the esophagus. The posterior vagal trunk has greater variability but is usually positioned between the posterior-medial wall of the esophagus and the right diaphragmatic crus. The anterior vagus nerve forms a hepatic division which separates from the anterior trunk at the level of the esophageal junction and passes between the avascular leaves of the gastrohepatic ligament. Fibers within the hepatic division innervate the gallbladder, biliary ducts, and liver.

The posterior vagal trunk gives rise to a celiac division which crosses the right diaphragmatic crus in parallel to the left gastric artery. The celiac division is usually

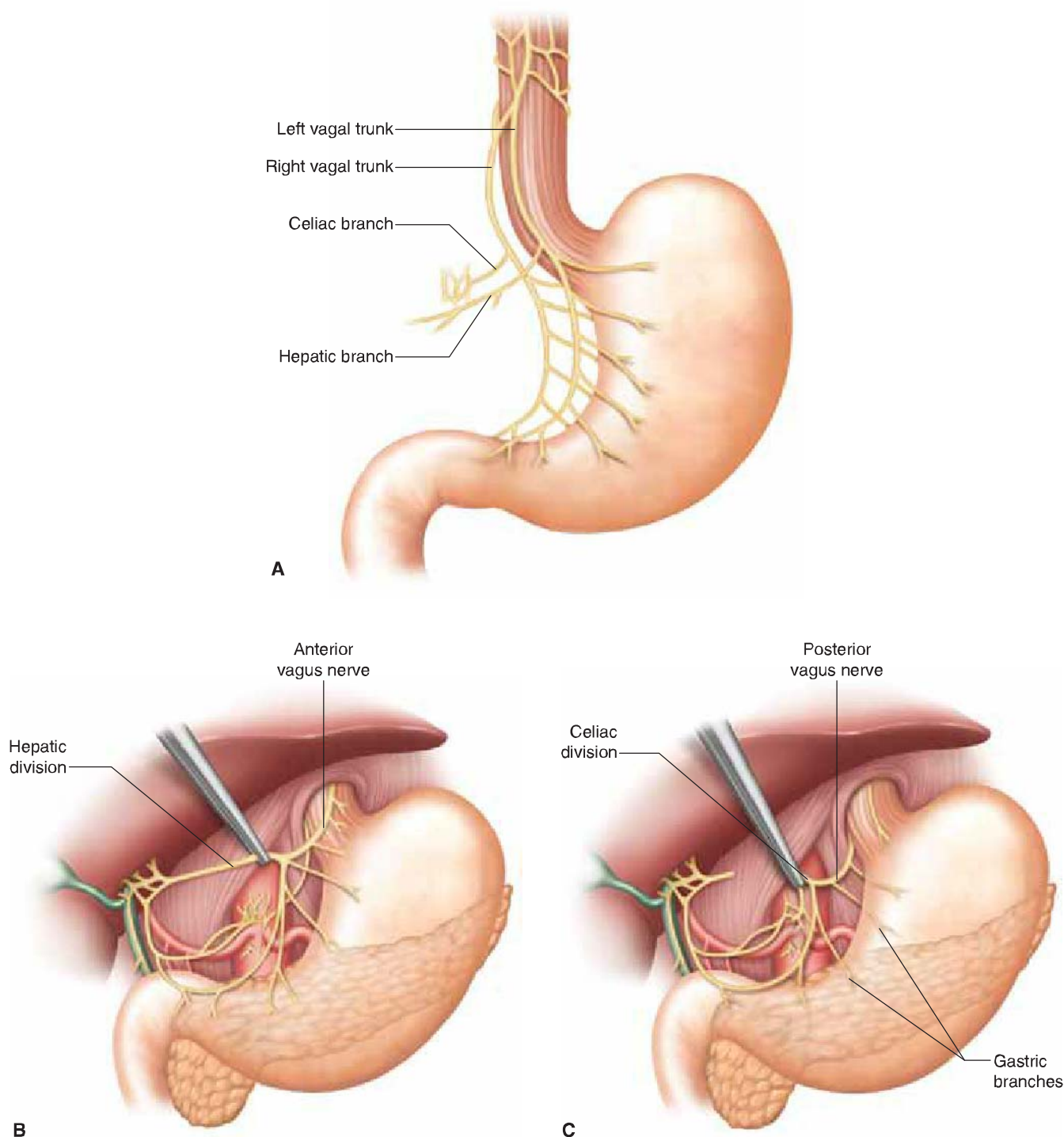
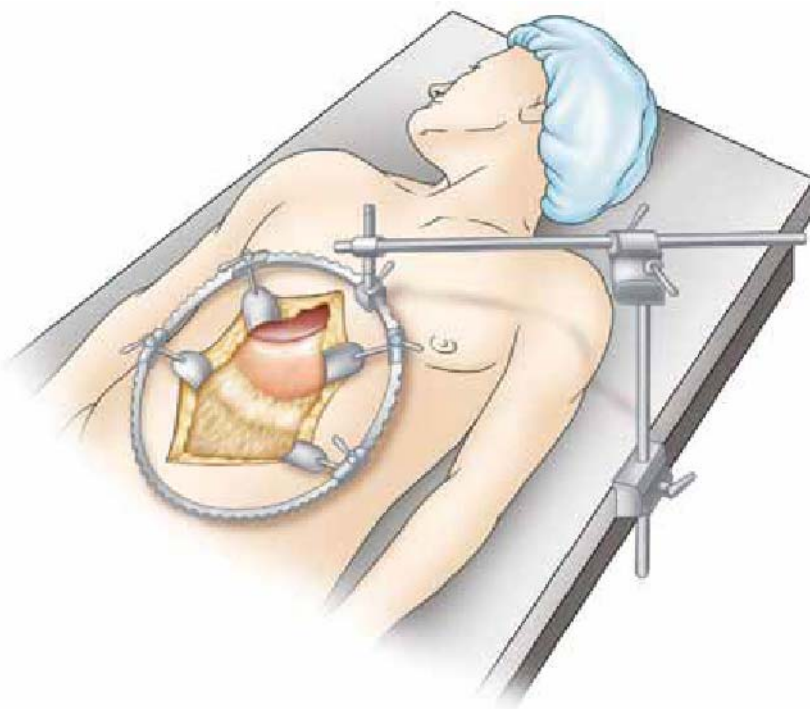


Figure 5.1 Vagal anatomy.

Figure 5.2 Topographic anatomy of the stomach.



larger than the hepatic division. Branches of the celiac division pass through the celiac ganglion. The anterior and posterior gastric divisions of the vagal trunks parallel the lesser curvature of the stomach and are contained within the gastrohepatic omentum. They run approximately 0.5 to 1 cm from the wall of the stomach along the lesser curvature. Small gastric branches terminate within the gastric fundus, body, and antrum (Fig. 5.1).

Position

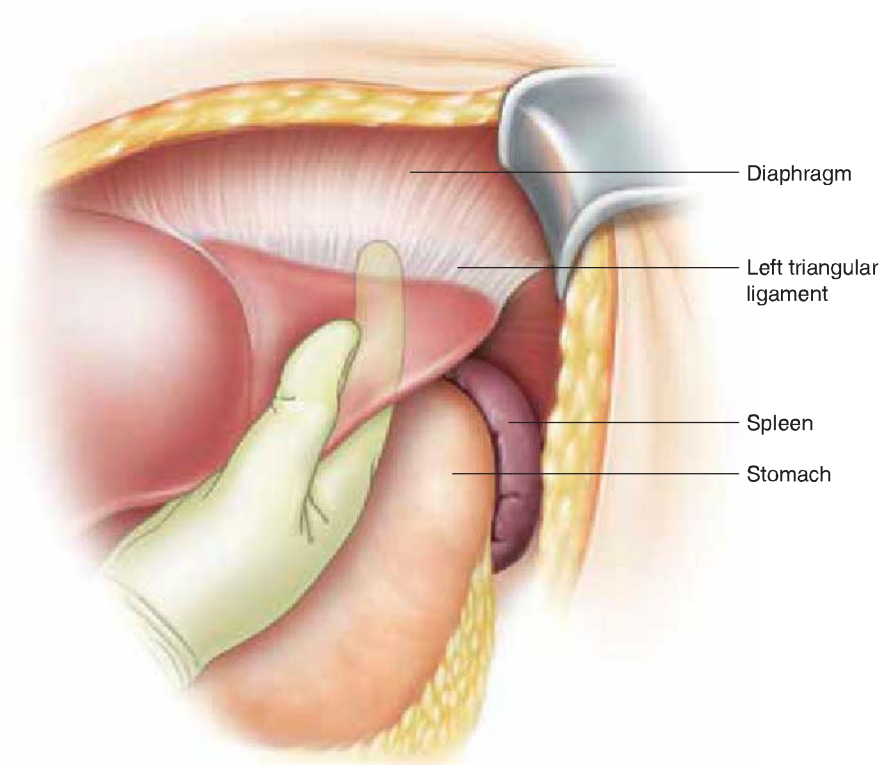
The patient is positioned supine on the operating table. A vertical midline incision extending from the xyphoid inferiorly provides exposure of the upper abdomen for truncal vagotomy performed via laparotomy. Exposure is provided by a self-retaining retractor. The costal margins should be retracted superiorly and elevated. A moderate amount of reverse Trendelenburg position is often helpful in shifting the omentum and other abdominal contents inferiorly and in improving exposure of the upper abdomen (Fig. 5.2).

Truncal Vagotomy

Exposure of the intra-abdominal esophagus can usually be obtained without mobilization of the left lateral segment of the liver. If the left lateral segment of the liver is hypertrophic or enlarged due to fatty infiltration, mobilization of this segment can sometimes improve exposure of the intra-abdominal esophagus. To do this, the surgeon places the right hand under the left lateral segment, palm upward, and retracts the liver inferiorly. The left triangular ligament is often thin and translucent. The fingers of the surgeon's hands are usually visible through the peritoneum (Fig. 5.3).

The left triangular ligament can be divided without cautery using this exposure. As the line of division proceeds medially, care must be taken to expose the anterior and posterior leaves of the triangular ligament separately. The peritoneal layers and the tissue between them may be divided individually using electrocautery. The inferior phrenic vein is often close to the posterior peritoneal edge of the triangular ligament,

Figure 5.3 Mobilization of the left lateral segment of the liver.



and care must be taken not to injure this vessel. This can be avoided by proper exposure. Mobilization of the left triangular ligament need not extend beyond the midline (Fig. 5.4).

With the left lateral segment of the liver retracted and protected, the anterior surface of the intra-abdominal esophagus can be appreciated visually and also by palpation of a previously placed nasogastric tube. The peritoneum overlying the esophagogastric

Figure 5.4 Division of the triangular ligament.

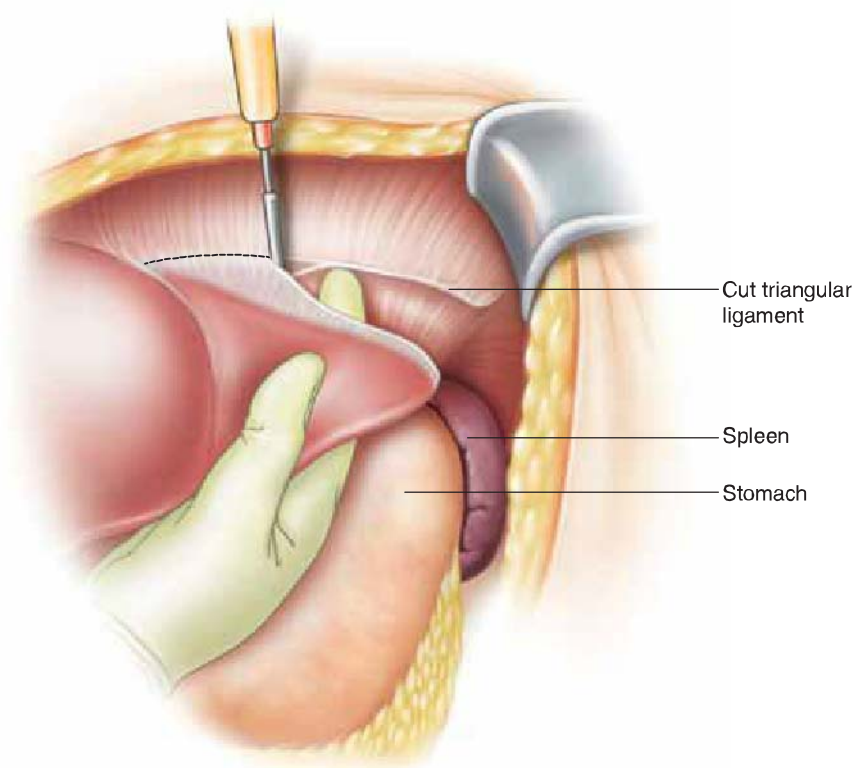
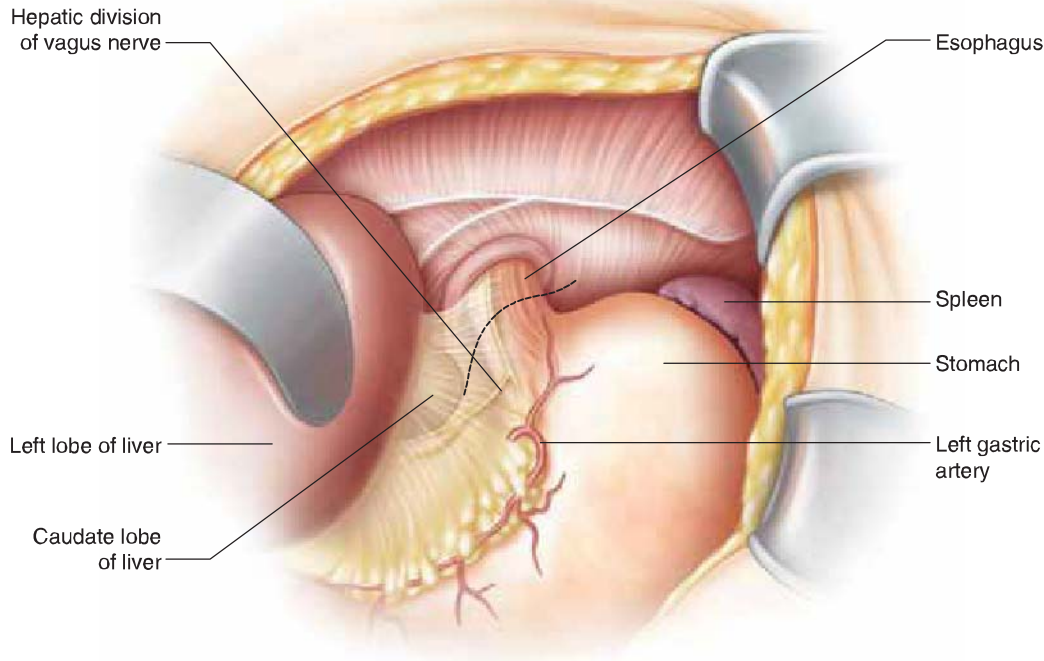


Figure 5.5 Exposure of the esophagogastric junction.

junction is divided transversely using electrocautery. In doing so, the hepatic branch of the anterior vagus nerve is sometimes visible in the leaves of the gastrohepatic ligament and should be preserved (Fig. 5.5).

After division of the peritoneum, the anterior vagus nerve can usually be seen on the anterior surface of the esophagus. The nerve can also be palpated with an index finger passed along the anterior surface of the esophagus. Retraction of the stomach caudally aids in the palpation of the vagal trunk by placing it on gentle tension (Fig. 5.6).

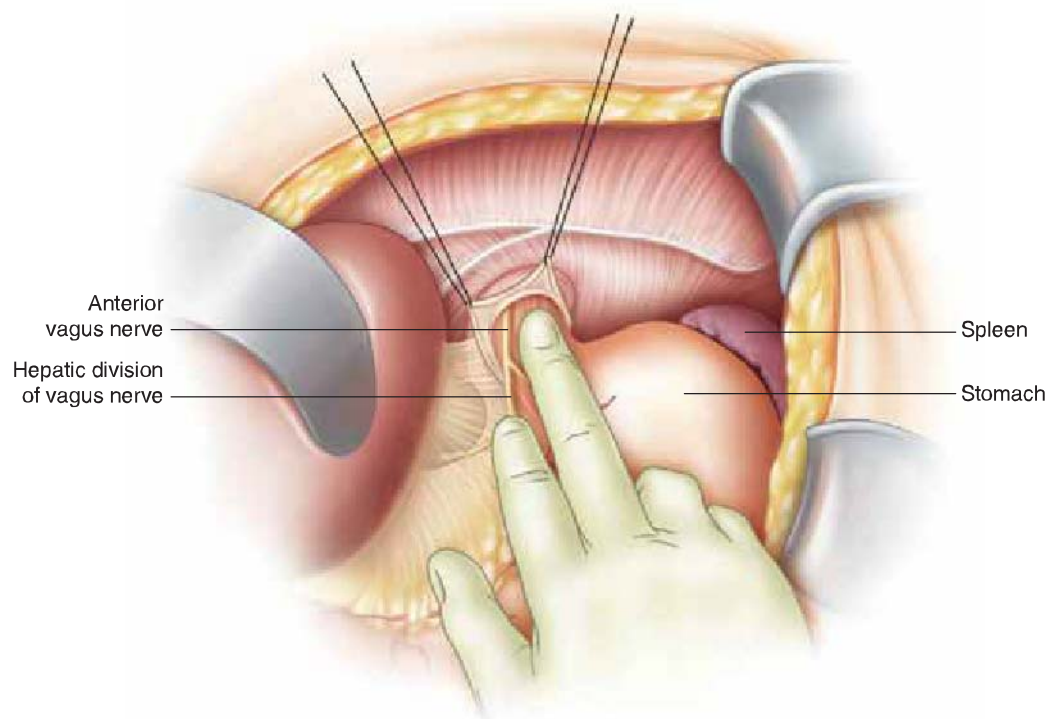
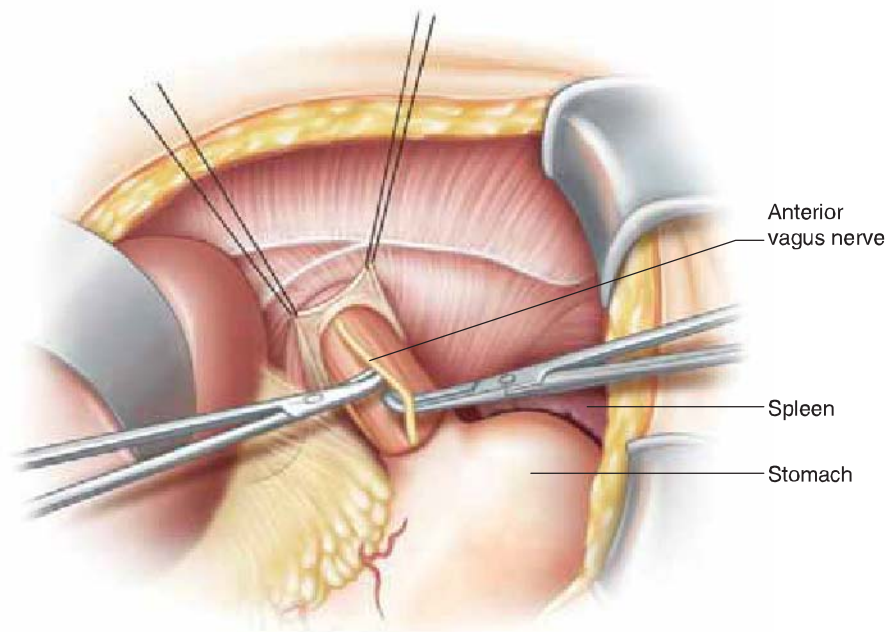
Figure 5.6 Exposure of anterior vagus nerve.

Figure 5.7 Isolation of the anterior vagal trunk.



The anterior vagal trunk is elevated from the anterior wall of the esophagus using a nerve hook or right angle. The nerve should be sharply dissected from the underlying esophageal musculature (Fig. 5.7).

The anterior vagus nerve is ligated both proximally and distally using metallic surgical clips. A segment between the clips is excised and sent for permanent pathologic examination (Fig. 5.8).

With the esophagus retracted to the patient's left, the operating surgeon then exposes the right diaphragmatic crus. The posterior vagal nerve can be identified visually or by palpation in the space between the posterior medial wall of the esophagus and the right crus. The position of the posterior vagal nerve can also be verified

Figure 5.8 Ligation of the nerve trunk.

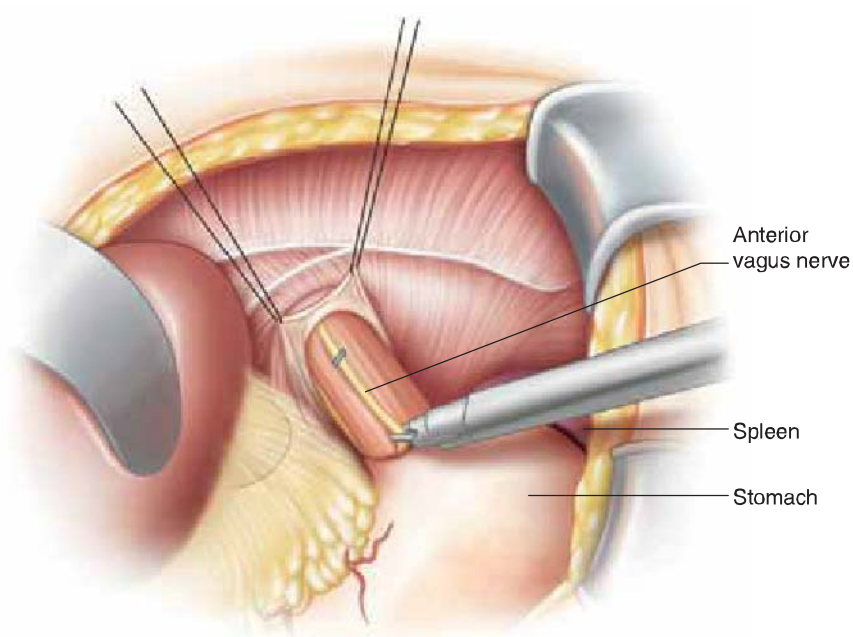
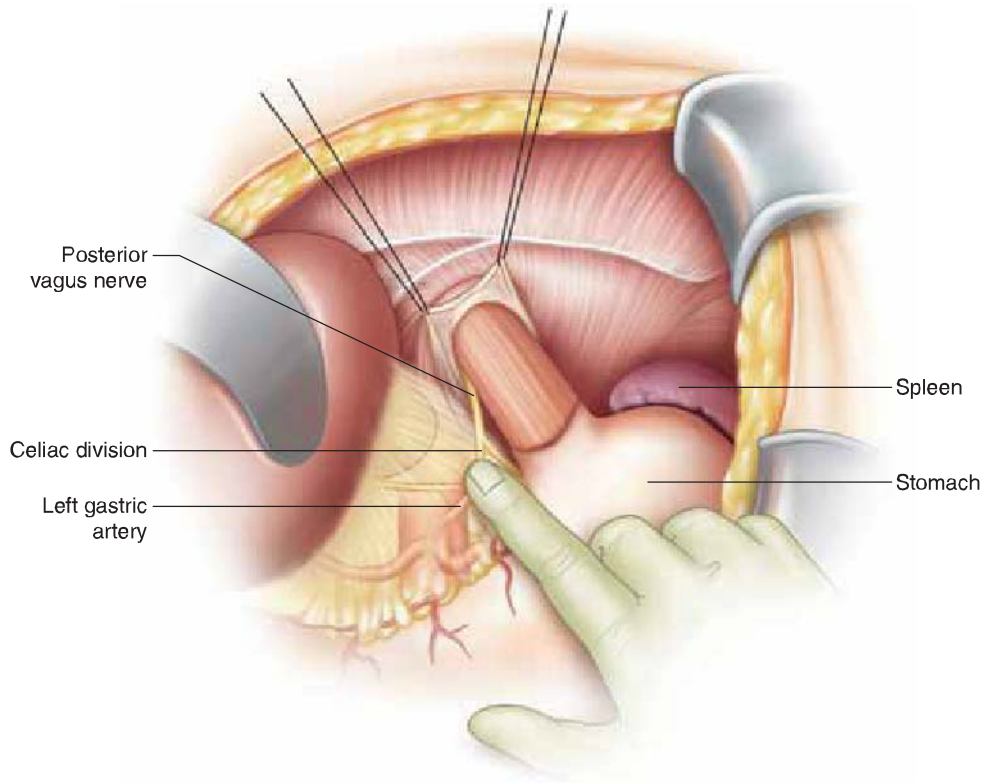


Figure 5.9 Exposure of the posterior vagal trunk.



by retraction of the celiac division; this maneuver places the posterior vagus nerve under some tension and accentuates its position. Because the celiac division of the posterior vagus nerve lies in proximity to the left gastric artery, this pulse can also be used as a landmark (Fig. 5.9).

The posterior vagus nerve is elevated from the surrounding tissue using a nerve hook or right angle clamp (Fig. 5.10).

A segment of the posterior trunk is excised after ligation by surgical clips (Fig. 5.11).

Figure 5.10 Isolation of the posterior vagal trunk.

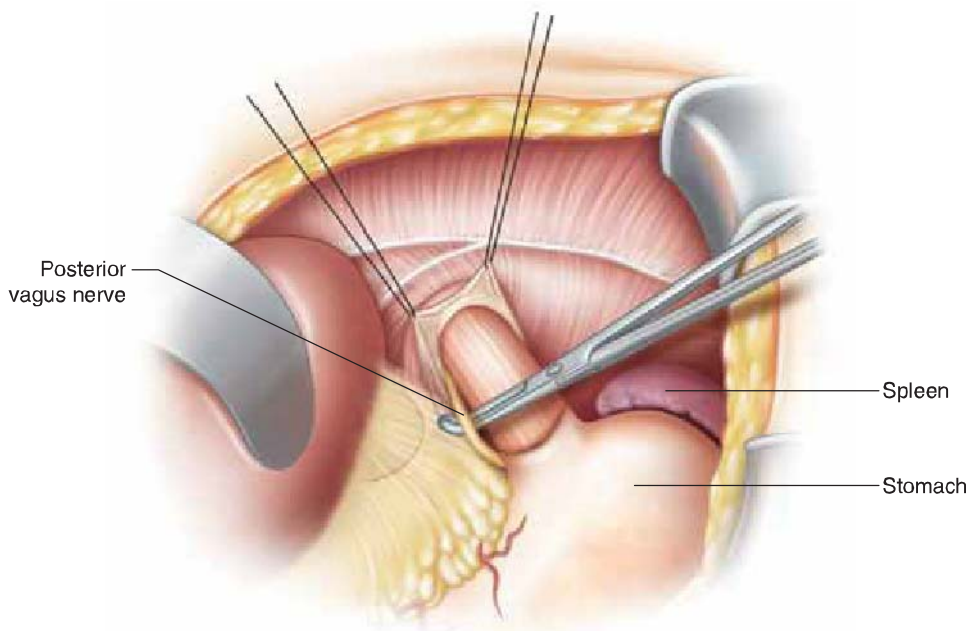
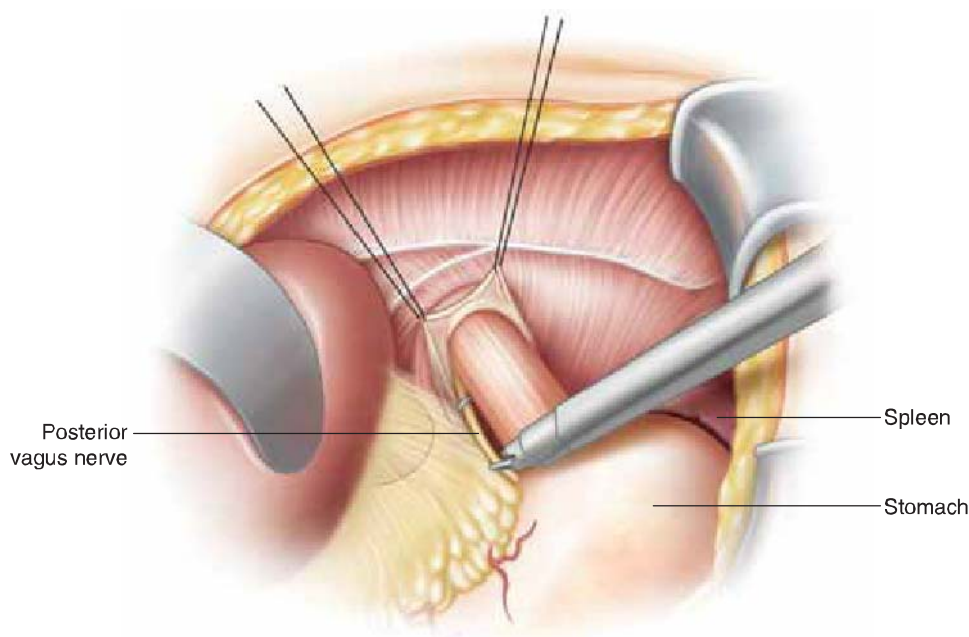


Figure 5.11 Nerve division.

Antrectomy

When operating for either benign or malignant disease, it is useful to examine the posterior wall of the stomach for possible inflammatory or neoplastic involvement of associated retrogastric strictures. This can be accomplished after entry into the lesser sac. The lesser sac is entered by dividing the attachment of the greater omentum to the transverse colon. An avascular plane can be developed immediately adjacent to the superior aspect of the transverse colon. This is best accomplished by retracting the omentum superiorly and retracting the corresponding segment of transverse colon inferiorly. While care must be taken not to injure the middle colic vessels, the omentum can be separated from the transverse mesenteric layers using electrocautery (Fig. 5.12).

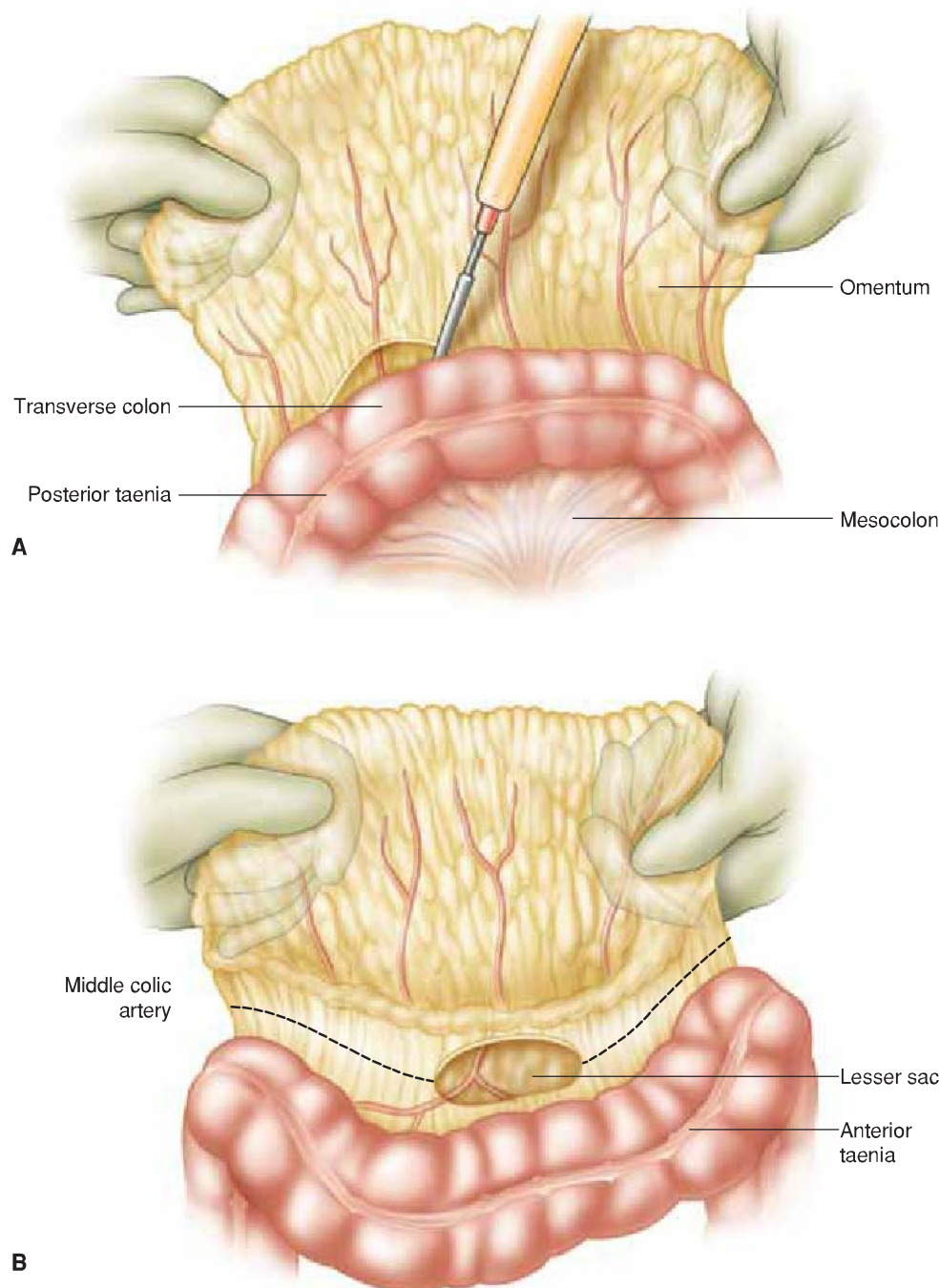
Alternatively, the retrogastric space can also be obtained by division of the avascular portion of the gastrohepatic omentum. Through this exposure, the posterior wall of the stomach and associated retrogastric strictures can be assessed (Fig. 5.13).

In treatment of neoplastic disease, the omentum may be resected en bloc with the distal stomach. Alternatively, for benign gastric diseases omental resection is not always necessary. The gastroepiploic vessels can be preserved. A line of dissection can be developed paralleling the left gastroepiploic vessel. Small blood vessels in this area can be taken between clamps and tied. In addition a number of alternative energy sources can be used during both open and laparoscopic operations to divide these vessels without suturing or ligatures (Fig. 5.14).

Antrectomy usually involves resection of approximately 50% of the distal gastric tissue. Preparation for a 50% gastric resection involves the division of the gastroduodenal ligament to a point approximately half way between the pylorus and the esophagogastric junction along the greater curvature. This point corresponds to the transition between the right and left gastroepiploic vessels. The gastric wall should be clear of fat and areolar tissue for a short distance in preparation for subsequent resection and then anastomosis (Fig. 5.15).

The right gastroepiploic vessels are exposed by retracting the stomach superiorly. The right gastroepiploic artery is divided close to its origin from the gastroduodenal artery. Fine areolar tissue between the posterior antropyloric region and the anterior surface of the pancreas can be divided sharply (Fig. 5.16).

Figure 5.12 Entry into the lesser sac. **(A)** The omentum is retracted superiorly and the transverse colon inferiorly. **(B)** An avascular plane above the transverse colon provides entry into the lesser sac.



The right gastric artery and its corresponding vein are often quite small. The right gastric vessels will be identified at the superior border of the proximal duodenum. The vessel should be divided close to the duodenal wall. Care must be taken to avoid injury to the common hepatic artery or common bile duct. Inflammation or tumor may obscure the anatomy in the region of the first portion of the duodenum (Fig. 5.17).

The area along the lesser curvature is used for division and anastomosis. Branches of the left gastric artery supply the lesser curvature. These are often represented as paired vessels to the anterior and posterior surfaces. These vessels should be ligated separately. Connective tissue and fat between these arterial branches and the gastric wall should be cleaned in preparation for subsequent anastomosis (Fig. 5.18).

The proximal duodenum is then divided using a GIA stapler. This provides closure at both ends and is convenient. The double staple closure of the divided duodenum

Figure 5.13 Incision of gastrohepatic ligament.

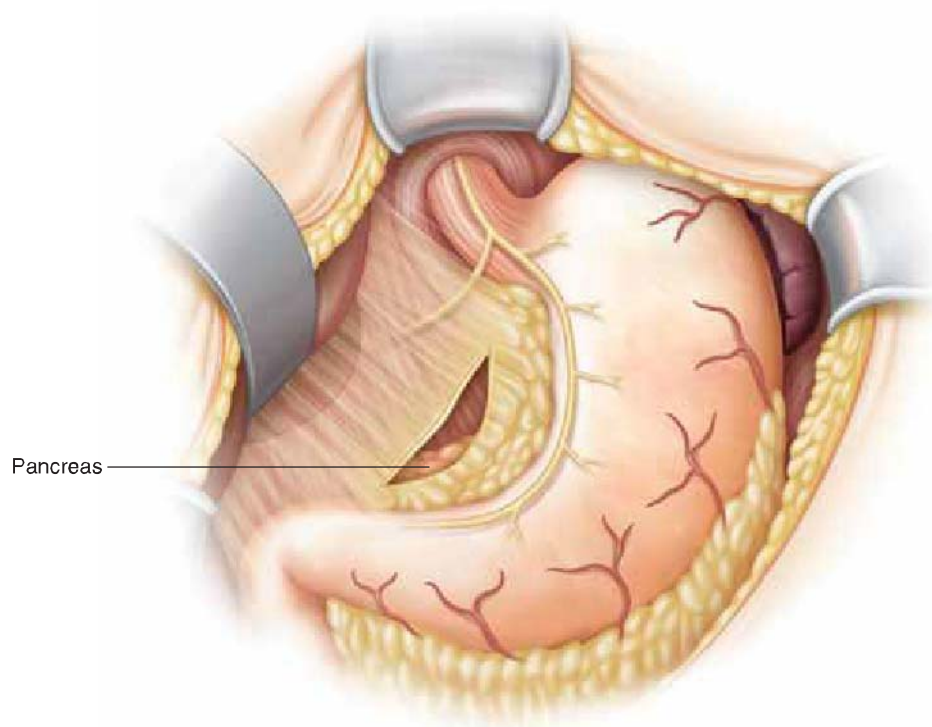
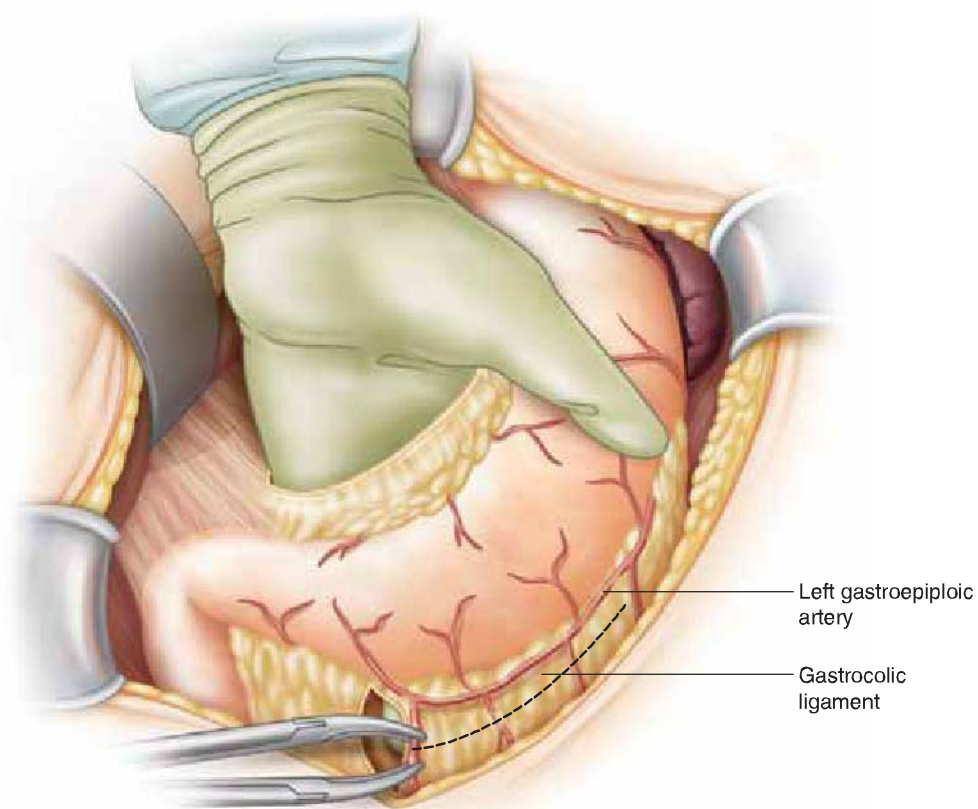


Figure 5.14 Division of gastrocolic omentum.



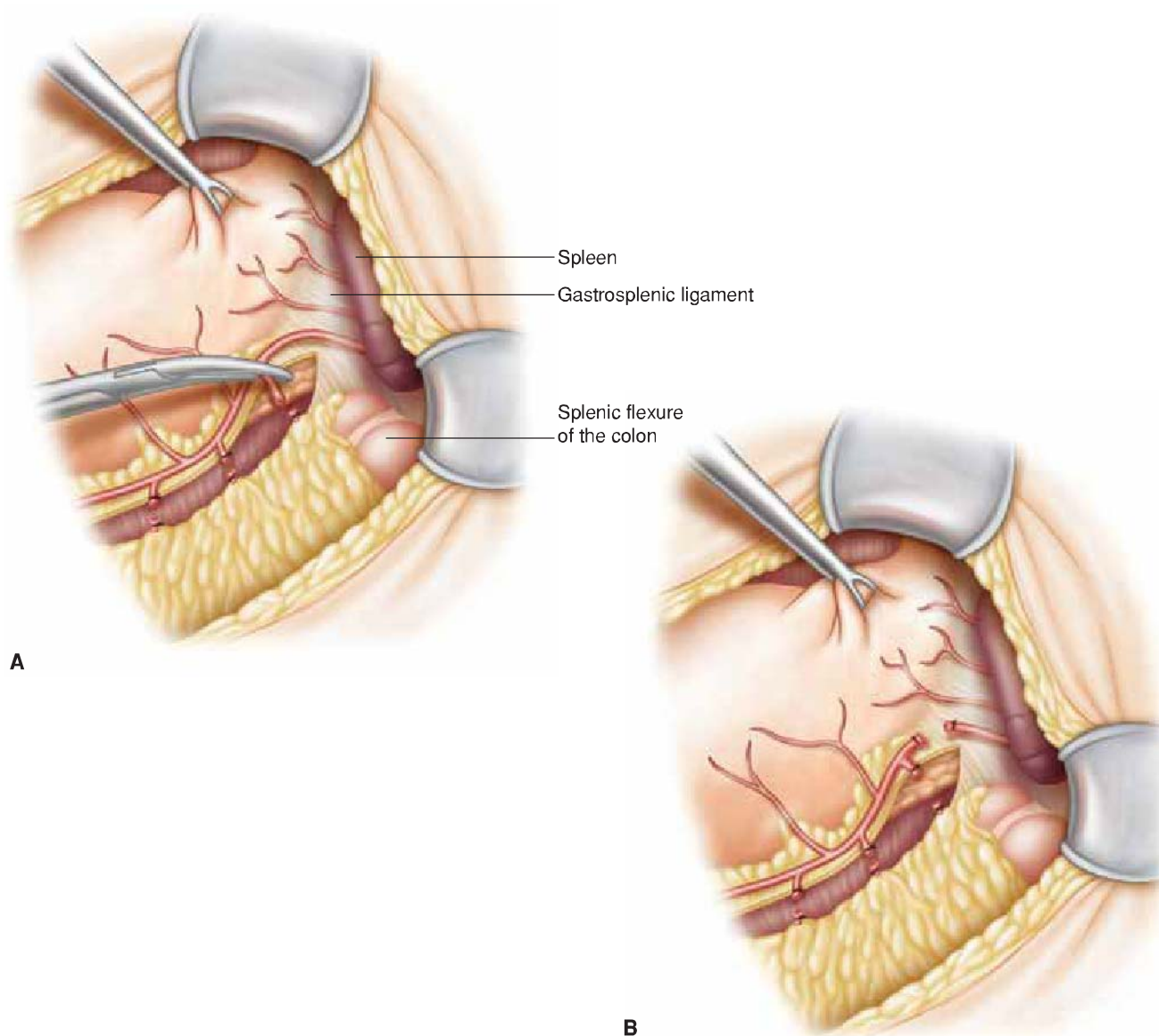


Figure 5.15 Preparation of greater curvature for hemigastrectomy. (A) Vessels entering the greater curvature of the stomach are individually ligated. (B) The dissection continues to the level of the short gastric vessels in the gastrosplenic ligament.

does not require suture reinforcement, although this can be done at the surgeon's preference using interrupted 3-0 nonabsorbable sutures (Fig. 5.19).

A TA-type stapler can be used to close the gastric remnant with a double row of staples. An occlusive clamp can be placed distally to prevent spillage of gastric contents during resection. Alternatively, this area may be divided with the application of a GIA-type stapling device (Fig. 5.20).

Billroth II gastrojejunostomy is illustrated as one of the means of restoration of gastrointestinal continuity. Billroth II reconstruction is most appropriate when the duodenum is not amenable to safe, tension-free anastomosis as a Billroth I reconstruction due to chronic scarring, inflammation, or inadequate length. A portion of the gastric wall is oversewn with interrupted sutures. A loop of proximal jejunum is brought through an incision in the transverse mesocolon to lie in tension-free apposition to the stomach. Interrupted nonabsorbable sutures are placed in a seromuscular fashion from the posterior gastric wall to the antimesenteric border of the jejunum. Using electrocautery, an incision is made in both organs of equal length. Gastrotomy is performed by partial excision of the stapled gastric closure. The posterior mucosal suture is created using continuous absorbable suture (Fig. 5.21).

Figure 5.16 Division of right gastroepiploic vessels.

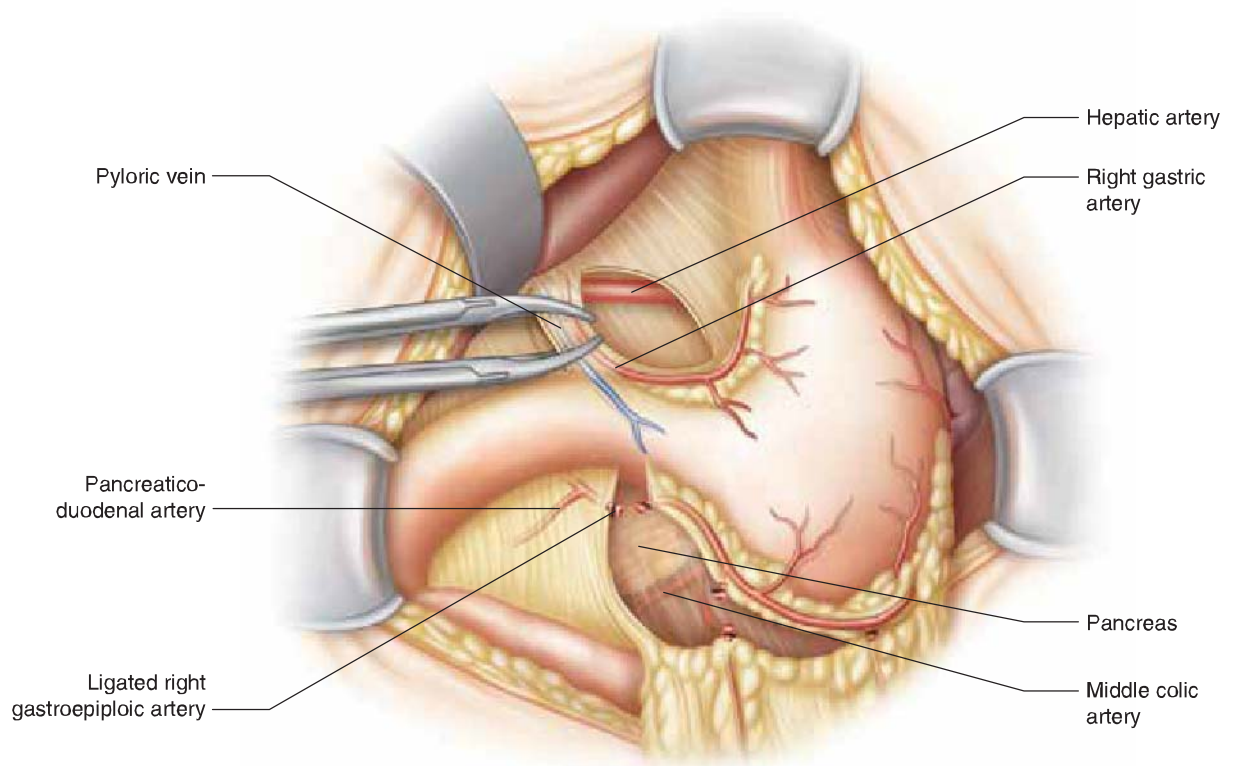
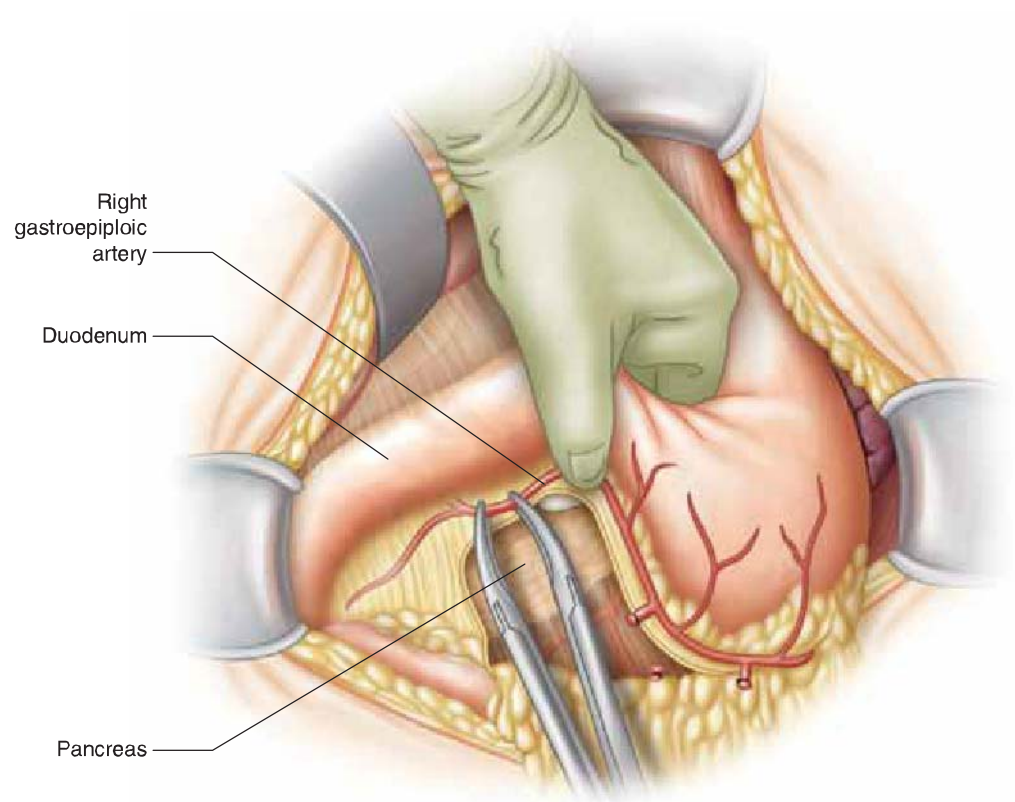


Figure 5.17 Ligation of right gastric vessels.

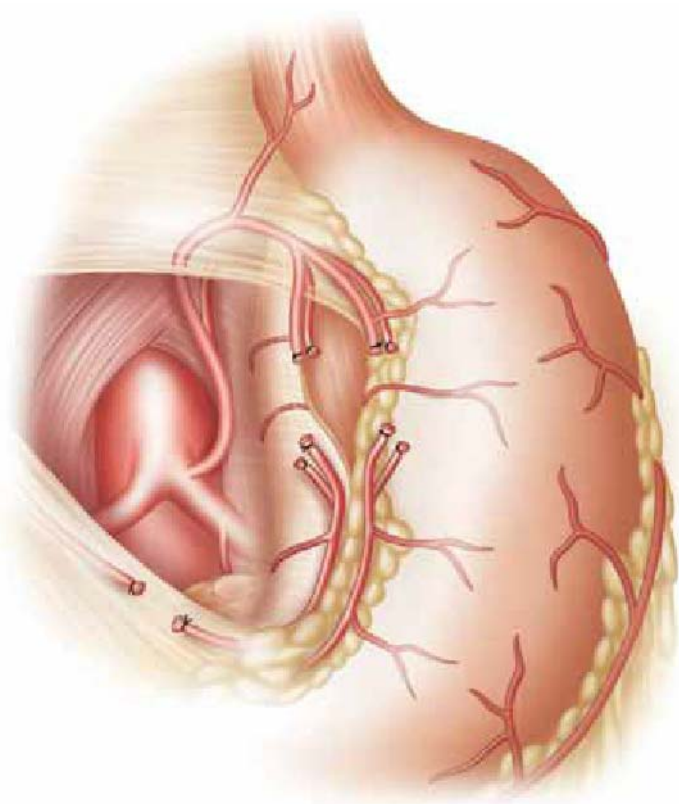


Figure 5.18 Dissection of lesser curvature.

This mucosal suture is continued along the length of the anterior aspect of the anastomosis. A layer of 3-0 nonabsorbable sutures is then placed in completion of the anterior anastomosis. Corner stitches must include the anterior gastric wall, posterior gastric wall, and the jejunum.

The gastrojejunal anastomosis is reduced inferior to the transverse mesentery. The anastomosis is secured in this position by sutures placed from the mesenteric defect to the gastric wall (Fig. 5.22).

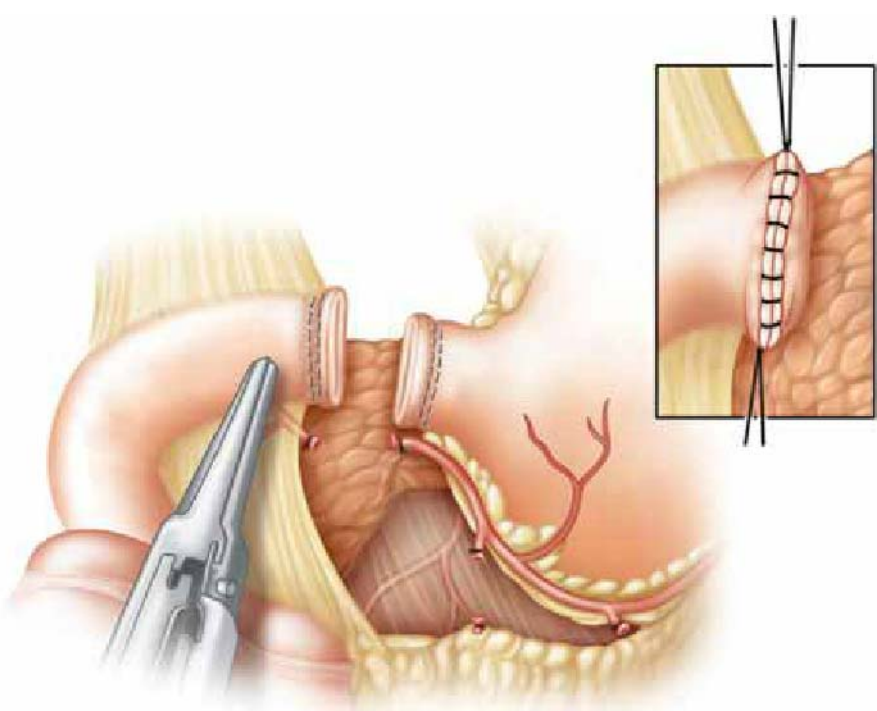
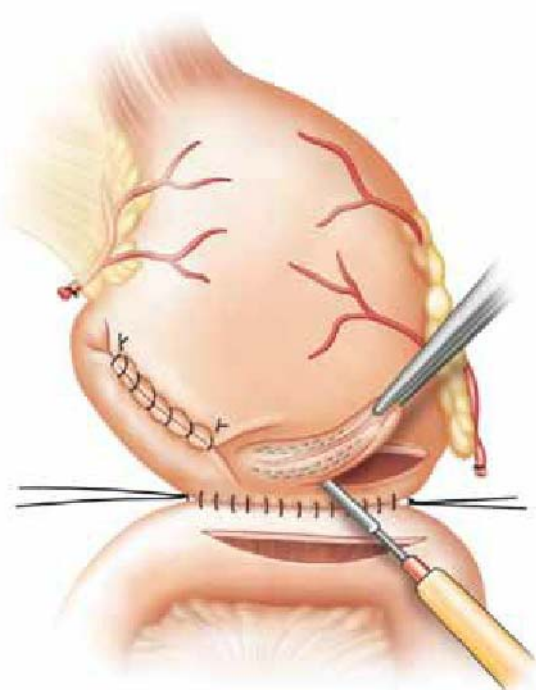
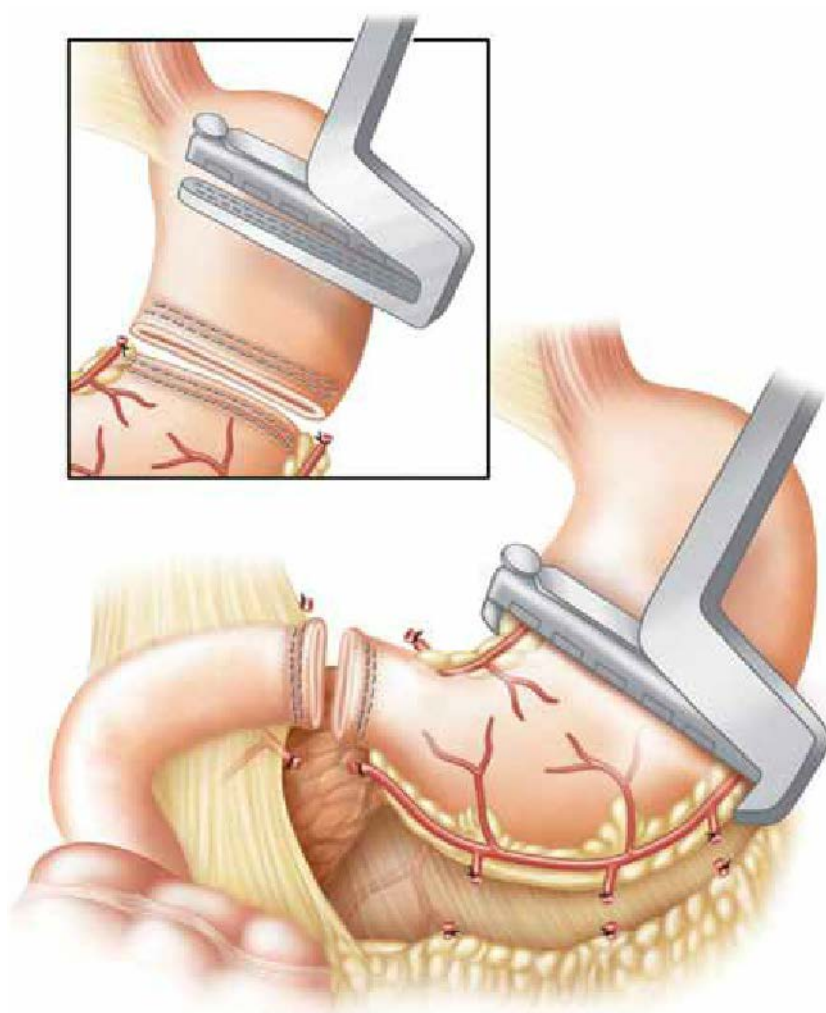
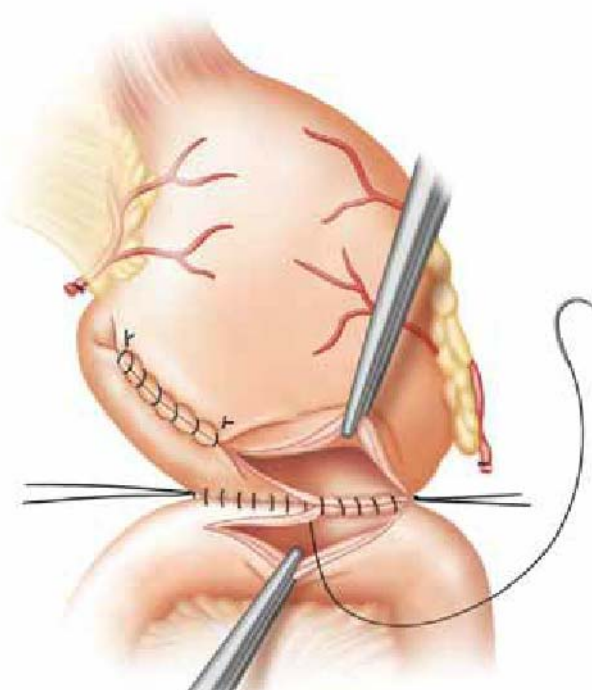


Figure 5.19 Division of the duodenum.

Figure 5.20 Division of the proximal stomach.



A



B

Figure 5.21 Gastrojejunostomy: sutured technique. (A) Posterior seromuscular sutures in place, matching incisions are made in the stomach and jejunum. (B) Beginning running mucosal suture.

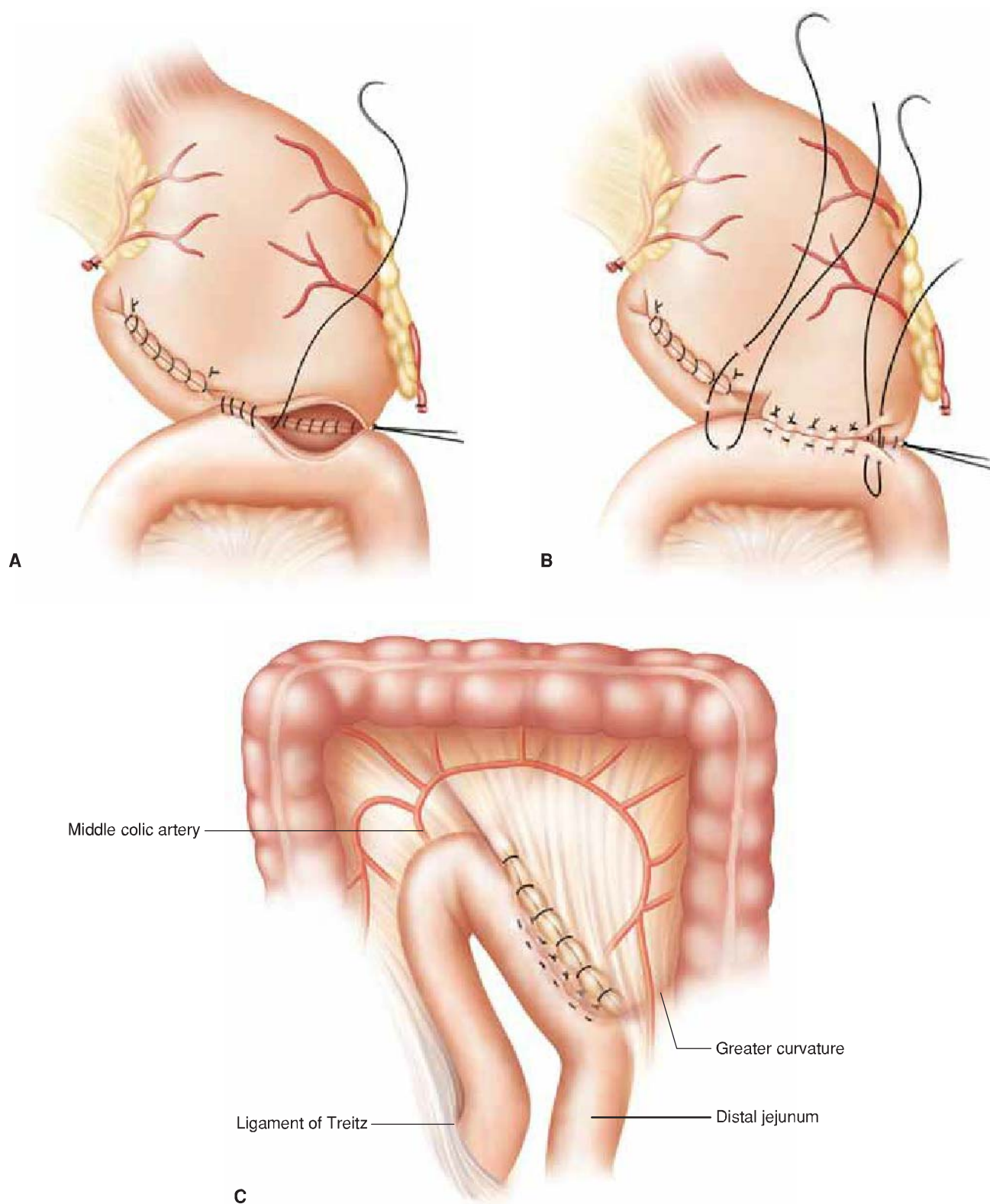


Figure 5.22 Anterior anastomosis. (A) Completion of running suture. (B) Anterior interrupted seromuscular suture line. (C) Delivery of gastrojejunal anastomosis inferior to the transverse mesentery and suture of the mesentery to the gastric surface.



POSTOPERATIVE MANAGEMENT

Nasogastric decompression is required for variable periods postoperatively. Gastric ileus or dysmotility is usually brief after elective operation; return of gastric emptying may be more prolonged if gastric outlet obstruction was present preoperatively. Atelectasis is more common after upper abdominal operations. Pulmonary hygiene and incentive spirometry are mandatory following antrectomy. If antrectomy has been performed for hemorrhage or in the setting of systemic hypotension, stress gastritis is a potential postoperative complication. The development of stress gastritis is critically dependent upon the presence of low intragastric pH; acid production suppression with proton pump inhibitors or histamine-2 receptor antagonists should be used as prophylaxis, in addition to attention to nutrition and restoration of normal mucosal perfusion.

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6 Laparoscopic Truncal Vagotomy with Antrectomy and Billroth II Reconstruction

Kfir Ben-David and George A. Sarosi Jr



INDICATIONS/CONTRAINDICATIONS

Recent advancements in pharmaceutical therapy and endoscopic treatments have significantly decreased the incidence of gastric surgery for peptic ulcer disease (PUD). With the advent of proton pump inhibitors, histamine receptor blockers, and multidrug therapy treatment for *Helicobacter pylori*, previously common operations for PUD have become very infrequent. Consequently, medical progress and new surgical innovations have completely transformed our approach to patients with benign gastric and PUD in need of surgical treatment. With the use of flexible endoscopy and minimally invasive surgical approaches the diagnosis and surgical treatment of PUD has resulted in less invasive surgical treatments resulting in shorter hospital stay, less overall cost, and quicker return to base-line activity while being able to maintain the same surgical technique without patient compromise.

The laparoscopic approach to PUD with truncal vagotomy, antrectomy, and Billroth II reconstruction is theoretically possible in nearly all patients with benign surgical disease who have failed medical and endoscopic treatments. Many minimally invasive gastrointestinal surgeons feel that PUD should be preferentially treated via a laparoscopic approach since it is associated with less intraoperative blood loss, earlier return of bowel function, less usage of analgesics, and a shorter postoperative hospital stay. Despite these strong case series favoring a laparoscopic surgical approach, there is a relative paucity of randomized clinical trials addressing this issue, likely due to the decreasing incidence of PUD and the infrequent need for elective surgical treatment. However, a number of patient characteristics represent relative indications and contraindications to the laparoscopic approach.

Even though the natural course of PUD is changing and intractable disease and pyloric obstruction as an indication for surgery is decreasing, the incidence of bleeding and perforation remains constant. Treatment of *H. pylori* and cessation of nonsteroidal anti-inflammatory drugs is imperative prior to any surgical procedure for PUD. Although there have been a number of case reports describing laparoscopic

surgical treatment with antrectomy, vagotomy, and Billroth II reconstruction for acutely perforated or bleeding PUD, many of these procedures were performed by very skilled laparoscopic surgeons. Furthermore, prior upper abdominal midline incisions and previous gastric and/or esophageal surgery are a relative contraindication to a laparoscopic truncal vagotomy, antrectomy, and Billroth II reconstruction. Mesh placement in the upper midline makes laparoscopic resection and reconstruction much more difficult. Liver cirrhosis and portal hypertension are also a contraindication to laparoscopic truncal vagotomy, antrectomy, and Billroth II reconstruction because of the risk of bleeding from gastric and esophageal varices. However, with increased experience utilizing laparoscopic approaches to gastric resection and reconstruction, patients with advanced age and its associated chronic medical conditions are not generally contraindications.



PREOPERATIVE PLANNING

Most of the preoperative evaluation is directed toward ensuring that the patient is adequately prepared for anesthesia. As a result, a careful assessment of the patient's fitness to undergo general anesthesia represents a major portion of the preoperative assessment. Because truncal vagotomy, antrectomy, and Billroth II reconstruction will almost always be performed under elective circumstances, chronic medical conditions such as cardiopulmonary disease and diabetes mellitus should be optimally managed prior to operation. The combination of careful history and documentation indicating the appropriate treatment of *H. pylori* and cessation of any nonsteroidal anti-inflammatory drugs usage is necessary to decrease the incidence of recurrent PUD. Additionally, a preoperative endoscopic evaluation of the gastric and duodenal mucosa is essential to rule out any abnormal pathology prior to surgical resection. Routine biopsy of nonhealing gastric ulcers is also imperative to exclude malignant disease. These can also be tattooed intraluminally preoperatively to help with intraoperative identification.

Confirmation of the correct diagnosis, ulcer location, previous surgical history, comorbidities, and patient's nutritional status are all important factors when treating patients with PUD with laparoscopic truncal vagotomy, antrectomy, and Billroth II reconstruction. Operative preparation should include a first- or second-generation cephalosporin in patients without achlorhydria or gastric outlet obstruction. Otherwise, a broader-spectrum antibiotic may be necessary for these patients. The intravenous antibiotic administration needs to be completed prior to skin incision and be discontinued 24 hours postoperatively. Prophylaxis for deep vein thrombosis can be achieved by the subcutaneous administration of heparin and the use of pneumatic compression devices prior to anesthetic induction, during the case and postoperatively.



SURGERY

After general anesthesia is administered, a bladder catheter is placed along with an orogastric or a nasogastric tube depending on the surgeon's preference. Although a nasogastric tube is often maintained postoperatively, routine gastric decompression has not been shown to affect outcomes in postgastrectomy patients. Depending on the surgical approach, the patient can be placed in supine or lithotomy position. For minimally invasive gastric resection approaches, our preferred method is to have the patient in a supine position allowing the operating surgeon to be on the right side of the patient while the assistant is on the contralateral side. The patient is secured to the table with two safety straps, a foot board, and all of their bony prominences are well padded. This positioning allows the patient to be securely placed in steep reverse Trendelenburg when performing the gastric resection and reconstruction. Lower and upper body warmer devices are applied to the patient throughout the case to help maintaining core body temperature.



Figure 6.1 Trocar and liver retractor placement.

The peritoneal cavity is accessed via a 5-mm port under direct visualization in the left subcostal region using a 5-mm 0° scope. The 5-mm 0° scope is then switched out to a 5-mm 30° scope. Three additional trocars are placed under direct visualization. A 5-mm trocar is placed in the supraumbilical region just left of the midline approximately 18–22 cm from the xiphoid process. A 12-mm trocar is placed on the contralateral side and a 5-mm trocar is placed in the right subcostal margin opposite the initial access trocar for the surgeon's right and left hand instruments, respectively. A 5-mm incision is also created in the subxiphoid region to allow for the placement of the Nathanson liver retractor (Fig. 6.1). This retractor is used to elevate the lateral segment of the left lobe of the liver and expose the gastroesophageal junction and anterior portion of the stomach.

The gastroduodenal omentum is dissected from the stomach permitting entry into the lesser sac. This is performed by cephalad retraction of the greater omentum while incising the avascular plane above the transverse colon (Fig. 6.2). The dissection continues at the pylorus with ligation of the right gastroepiploic artery using a laparoscopic vascular stapler inserted through the 12-mm trocar (Fig. 6.3). This dissection proceeds along the greater curvature of the stomach and ends halfway between the pylorus and the gastroesophageal junction. This maneuver spares the left gastroepiploic vessels and the short gastric vessels. If there is any concern as to the level of division of the stomach to achieve adequate margins an intraoperative endoscopy can help determine the site of gastric resection. The posterior wall of the stomach is separated from the anterior pancreas and base of the transverse mesocolon by blunt dissection and sharp division of connective tissue attachments which can be very inflamed and dense in some patients. The duodenum is carefully Kocherized, and in patients with pyloric inflammation, care must be taken to avoid injury to the bowel, common hepatic artery, common bile duct, and portal vein. The right gastric vessels are similarly ligated close to the stomach using a laparoscopic vascular stapling device or ultrasonic shears (Fig. 6.4).

The gastrohepatic ligament is incised, and the lesser curvature is dissected. The right crus is identified and dissected along the medial border in a cephalad fashion up to the diaphragm and circumferentially to the left crus. Using a pars flaccida technique,



Figure 6.2 Gastrocolic omentum dissection.

a Penrose is placed around the GE junction through the retroesophageal space. The phrenoesophageal ligament is divided as the stomach is retracted in a caudal posterior fashion. Meticulous and direct dissection will prevent injury to the spleen, short gastric vessels, esophagus, or stomach while performing this maneuver. A minimum of 5 cm of the esophagus is mobilized to allow adequate space to identify and ligate the vagus nerves. This dissection exposes the distal esophagus along with the anterior and the posterior trunks of the vagus nerves. Each nerve is individually identified, freely dissected from the esophagus, and excised. Approximately 1 cm of fiber from the anterior and posterior trunk of the vagus nerves is excised and sent to pathology for histologic confirmation (Fig. 6.5).

The proximal duodenum is divided with care with a 60-mm endoscopic linear stapler to avoid injury to the common hepatic artery, common bile duct, and portal vein (Fig. 6.6). The duodenal staple line can be reinforced with interrupted imbricated sutures or stapler buttressing at the discretion of the surgeon. Once the proximal duodenum is divided the distal stomach is rotated toward the patient's left side to help assist with the proximal transection of the stomach. Similarly, the proximal end of the stomach is divided with multiple linear stapling devices from the lesser to the greater

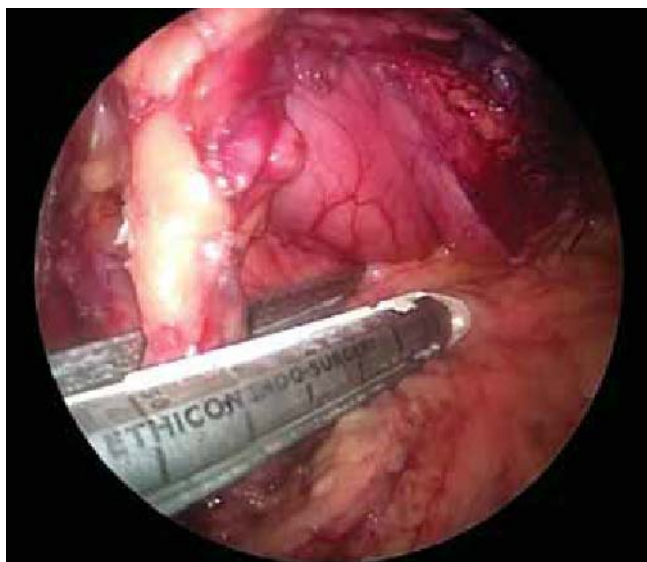


Figure 6.3 Ligation of right gastroepiploic vessels.



Figure 6.4 Division of right gastric vessels.

curvature via the right-sided 12-mm trocar (Fig. 6.7). The resected stomach is moved to the right upper quadrant and is removed via an endoscopic specimen bag. In order to reduce problems with loss of pneumoperitoneum from the extraction site the specimen is not removed until the remainder of the case is completed.

The laparoscopic Billroth II reconstruction can be performed by various surgical techniques. The authors prefer a retrocolic, sutured, and linear stapled gastrojejunostomy. This portion of the operation requires the patient to be in and out of reverse Trendelenburg position to accommodate appropriate positioning for the reconstruction. A proximal jejunal loop 30 to 40 cm distal to the ligament of Treitz is delivered through an incision in the transverse mesocolon (Fig. 6.8). The jejunum is brought up in cephalad fashion via the transverse mesocolon defect to the newly created gastric pouch in an isoperistaltic manner (Fig. 6.9). The antimesenteric surface of the jejunum is apposed to the posterior gastric wall starting from the greater to the lesser curvature of the stomach with a nonabsorbable suture in a continuous fashion (Fig. 6.10). Matching gastrotomy and enterotomy are performed with electrocautery or ultrasonic shears. The laparoscopic stapler's cartridge and anvil limbs are inserted into the gastrotomy and enterotomy incisions, respectively (Fig. 6.11). After the stapler is fired, the device is withdrawn and the staple line is inspected for hemostasis. The common gastrotomy and enterotomy defect is closed with the application of a laparoscopic stapling device



Figure 6.5 Excision of Vagus Nerve.

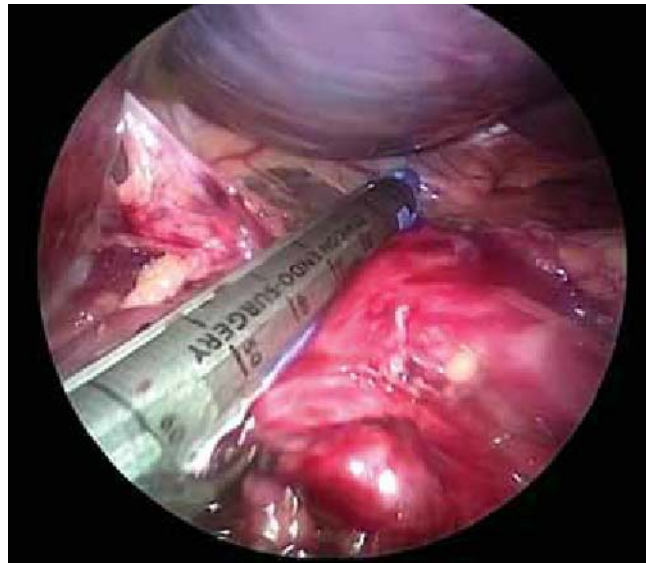


Figure 6.6 Division of proximal duodenum.

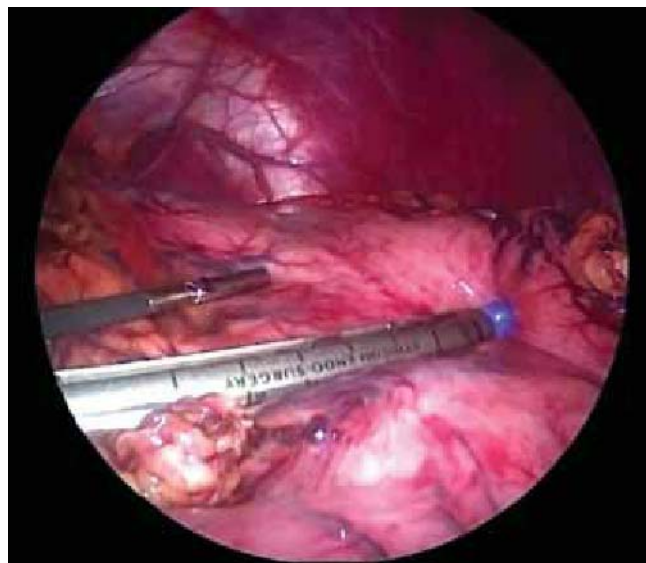


Figure 6.7 Gastric resection using linear stapler.



Figure 6.8 Transverse mesocolon window.

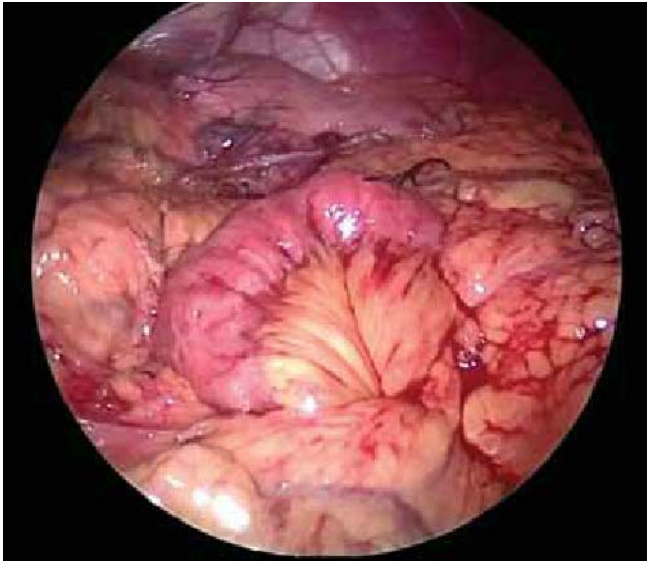


Figure 6.9 Jejunum via transverse mesocolon defect in an isoperistaltic manner.



Figure 6.10 Jejunum is apposed to the posterior gastric wall.

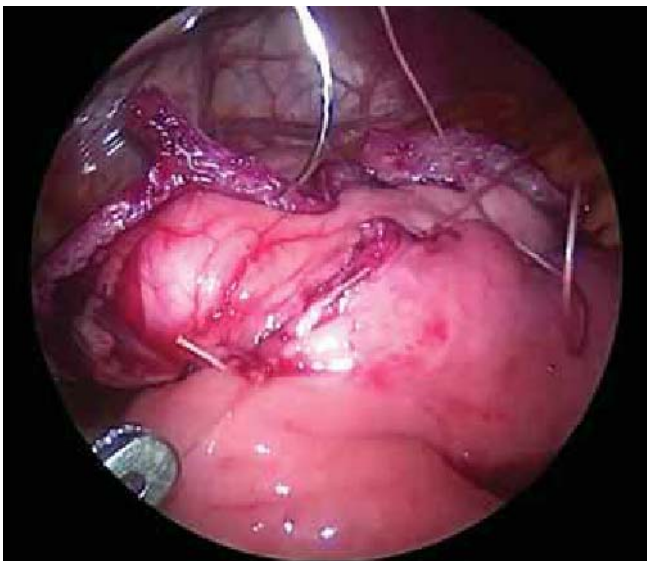


Figure 6.11 Closure of common gastrotomy and enterotomy defect.

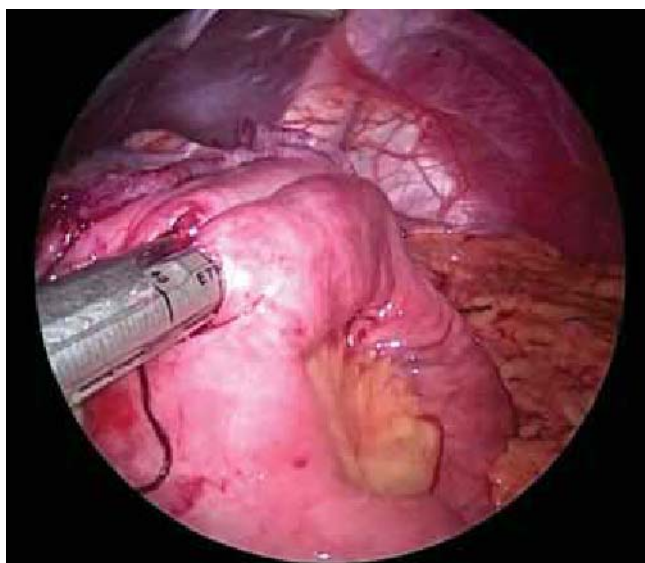


Figure 6.12 Linear stapler's cartridge and anvil limbs are inserted into the gastrotomy and enterotomy.

or with a continuous absorbable suture (Fig. 6.12). The entire staple line and/or suture line is inverted in a running fashion using a nonabsorbable suture (Figs. 6.13 and 6.14). The transverse mesocolon defect is reapproximated using nonabsorbable interrupted sutures. The jejunal loop is also sutured in an interrupted fashion to the transverse mesocolon defect to prevent potential bowel herniation through this defect (Fig. 6.15). A flexible endoscope is frequently used in these cases to determine the adequacy of the gastric inlet and/or outlet, visualization of the intraluminal staple line, and evaluation of the integrity of the anastomosis under saline irrigation (Fig. 6.16). Finally, the resected stomach is placed in an endoscopic specimen bag and retrieved via the 12-mm trocar. The 12-mm trocar fascial incision site is closed. The remaining 5-mm trocars and Nathanson liver retractor are removed under direct visualization. Local anesthetic is administered to all incision sites. The subcutaneous tissue and skin are reapproximated with absorbable subcuticular suture and Dermabond dressing, respectively.

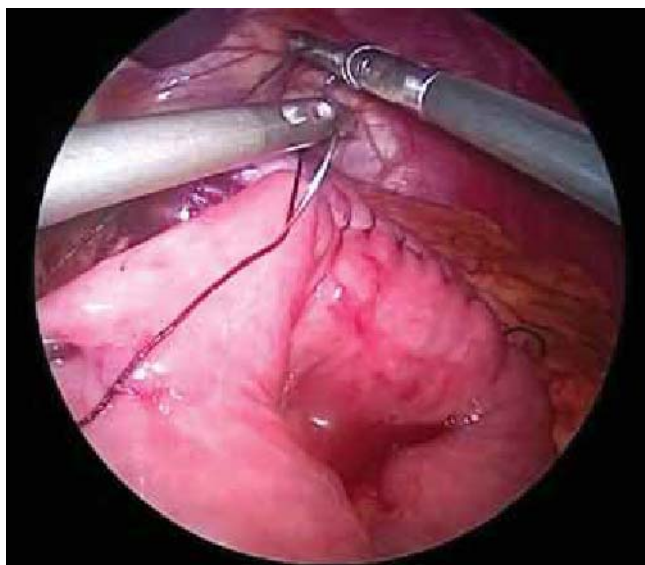


Figure 6.13 Inversion of stapled and suture line.

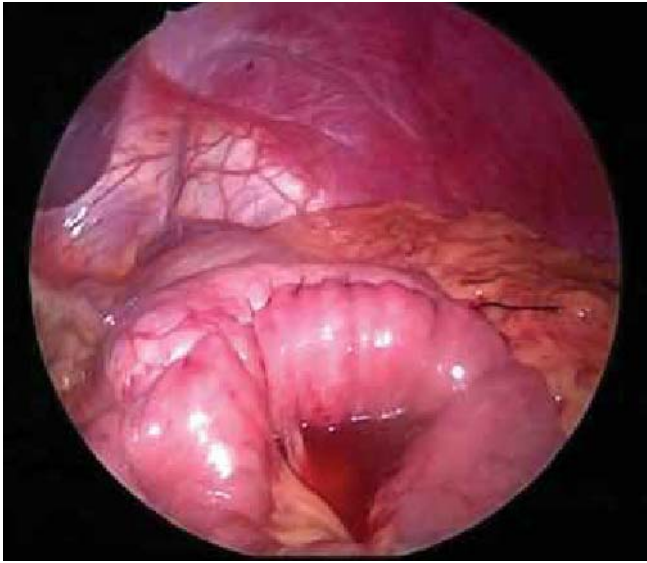


Figure 6.14 Completed anastomosis.

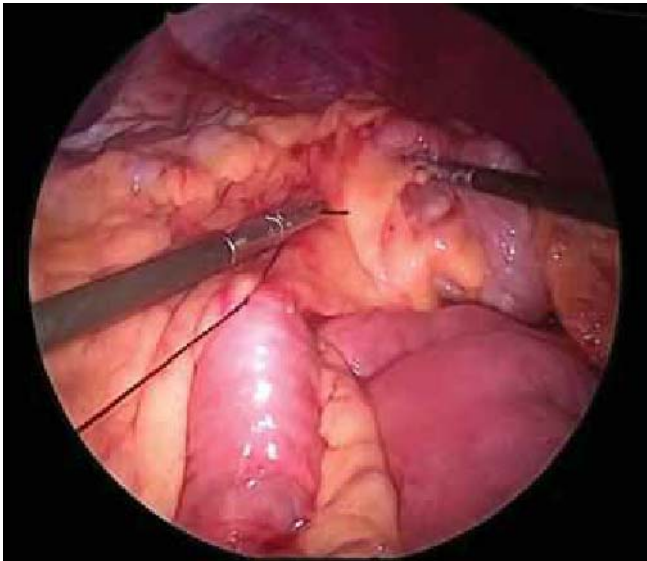


Figure 6.15 Jejunal loop sutured to the transverse mesocolon defect.

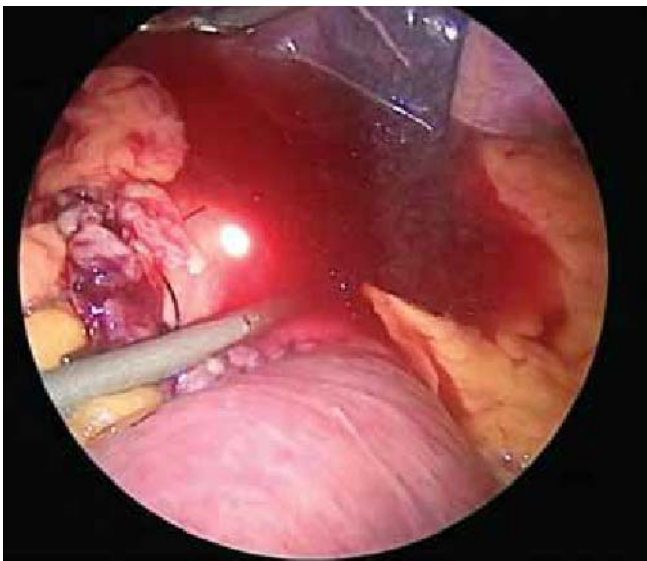


Figure 6.16 Endoscopic evaluation of anastomosis.



POSTOPERATIVE MANAGEMENT

All patients are admitted to our surgical floor and are encouraged to ambulate on the same day of surgery. Deep venous thrombosis prophylaxis is continued during their hospitalization with subcutaneous heparin injections and sequential compression devices while in bed. The patient's urinary catheter and antibiotics are both discontinued on postoperative day 1. Pain is controlled with intravenous narcotic medication for the first 24 hours followed by oral analgesic medication. The majority of the patients are started on a clear liquid diet on the first postoperative morning. In patients without adynamic small intestinal ileus their diet is quickly advanced as tolerated. Most patients are discharged home on postoperative day 3 and instructed to eat 4 to 6 small meals a day. Prior to discharge, all patients are evaluated by our registered dietitian and are instructed about a postgastrectomy diet. All patients return for their routine postoperative visit 2 weeks following hospital discharge to ensure appropriate healing of their incisions and adequate nutritional intake.



COMPLICATIONS

As with most laparoscopic operations, the patient's postoperative course is usually benign but any deviation from the normal course requires further investigation as with all other surgical patients. With meticulous and hemostatic dissection, bleeding after laparoscopic surgery is fairly uncommon but can occur from the trocar site insertions, multiple vessel ligations, transected stomach, or intraluminal anastomosis. These are often identified intraoperatively by careful inspection at the completion of the case with the aid of the laparoscope and a flexible endoscope.

A number of postgastrectomy syndromes have been described following gastric operations performed for PUD and gastric neoplasms, regardless of the surgical approach, which can be disabling in 1% to 2% of patients. The two most frequent manifestations are dumping and alkaline reflux gastritis. Although the precise cause of dumping syndrome is not known, it is believed to be related to the rapid passage of ingested food into the proximal small bowel. Approximately 10% to 15% of patients who have had a truncal vagotomy, antrectomy, and Billroth II reconstruction will experience mild dumping symptoms of postprandial nausea, epigastric pain, diarrhea, and palpitations. Minor dietary alterations and passage of time will improve the symptoms for most patients. In the most severe of cases, low-dose subcutaneous octreotide, a somatostatin analogue, can be administered to help with early and late dumping symptoms.

Alkaline reflux gastritis is often associated with nausea and emesis, evidence of bile reflux in the stomach, and histological evidence of gastritis. Patients who present with this syndrome must undergo endoscopic examination to exclude recurrent ulcer. Unfortunately the only proven treatment is the conversion of gastrojejunostomy to Roux-en-Y configuration to prevent reflux of intestinal contents and bile.

Despite these well-described postoperative complications, ulcer recurrence rate is the lowest for patients with truncal vagotomy and antrectomy regardless of surgical technique. With the laparoscopic approach, the patient's ability to return to normal daily activities and work, incidence of wound infections, hospital analgesic use, and hospital length of stay are all improved when compared to open truncal vagotomy, antrectomy, and Billroth II reconstruction.



RESULTS

Although there are few large series reporting the outcomes after laparoscopic vagotomy and antrectomy for intractable or obstructing peptic ulcers, experience from the use of open vagotomy and antrectomy suggests that excellent long-term outcomes can

be achieved. Multiple authors have reported ulcer recurrence rates of 1% to 2% percent after antrectomy and vagotomy. Rivera and colleagues in a series of 14 patients who underwent laparoscopic gastrectomy for ulcer disease reported a conversion rate of 5%, a mean operative length of 165 minutes with a mean estimated blood loss of 84cc. Time to resumption of diet was 3 days and mean length of stay was 4 days. Complications were seen in 14% percent of patients. Goh and colleagues reported the outcomes of 63 laparoscopic gastric resections for ulcer disease. They noted a conversion rate of 5% with a mean operative length of 215 minutes. The mean time to resumption of liquid diet was 3 days, solid diet 5 days, and mean length of stay was 8 days. Complications were observed in 13% of patients. The mean time to return to usual activities was 17 days.



CONCLUSIONS

Laparoscopic truncal vagotomy with antrectomy and Billroth II reconstruction is a technically challenging but feasible operation in the patient with intractable peptic ulcer or gastric outlet obstruction secondary to PUD. In skilled hands it can be performed with outcomes comparable to open truncal vagotomy and antrectomy and offers the potential to reduce postoperative hospitalization and recovery time.

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7 Truncal Vagotomy with Antrectomy and Roux-En-Y Reconstruction

Kevin E. Behrns and Angel M. Caban



INDICATIONS/CONTRAINDICATIONS

The introduction and evolution of acid-suppressive medications and the discovery of *Helicobacter pylori* as the cause of peptic ulcer disease have decreased markedly the number of patients who require elective surgical therapy. However, even though surgical therapy is indicated only in refractory cases of ulcer disease, it is essential that practicing general surgeons understand the pathophysiology and surgical options for the treatment of peptic ulcer disease. The early and important contributions of Dr. Lester Dragstedt identified the vagus nerve as an important stimulus to acid secretion. His experimental work provided the rationale for vagotomy as a mechanism to eliminate the direct cholinergic stimulus for acid secretion to the parietal cells. His work set the stage for the development of three variations of surgical vagal stimulus interruption: truncal vagotomy, selective vagotomy, and highly selective vagotomy. Since truncal vagotomy markedly alters gastric motility patterns, a gastric drainage or emptying procedure should be performed in conjunction with truncal vagotomy. Several surgical options for gastric drainage exist, but this chapter will focus on truncal vagotomy with antrectomy and Roux-en-Y reconstruction. It should be recognized that this operation was developed experimentally by Cesar Roux as a model of gastrojejunal anastomotic ulcer and, therefore, should be used only in refractory cases of ulcer disease. In fact, many surgeons preferably perform a gastroduodenostomy (Billroth I) or a gastrojejunostomy (Billroth II) for reconstruction after antrectomy because these operations represent more physiologic reconstructions of gastroenteric continuity.

Despite the limited application of this procedure, the indications for surgical intervention have not changed; surgery is often needed for intractable ulcers, bleeding, perforation, or obstruction. Importantly, the rate of emergency operations for complicated peptic ulcer disease has not changed since the introduction of acid-suppressive medications and antibiotics. Thus, emergent surgical therapy for complicated ulcer disease remains constant and requires that the general surgeon be up-to-date regarding new

advances in surgical treatment. However, the emergent management of complicated peptic ulcer disease is beyond the scope of this work. The aims of this chapter are to describe the modern surgical approach to truncal vagotomy and antrectomy with Roux-en-Y reconstruction with emphasis on the indications, techniques, and postoperative outcomes. It is important to realize that the treatment of peptic ulcer disease is often individualized on the basis of the location of the ulcer, the inflammatory response, and the mode of presentation.



PREOPERATIVE PLANNING

Truncal vagotomy and antrectomy with Roux-en-Y reconstruction is most often performed in an elective setting for recalcitrant peptic ulcer disease, and hence, a complete upper gastrointestinal tract evaluation should be completed. The evaluation should include an upper flexible endoscopy to identify the location of the ulceration, to rule out malignancy, and to test for *H. pylori*. In patients with prior upper gastrointestinal surgery a barium contrast study or computed tomography with oral contrast may be useful in delineating surgical anatomy. Serum gastrin level should also be obtained in patients with recurrent ulcers that are refractory to medical therapy and salicylate concentration should be determined for patients on nonsteroidal anti-inflammatory medications. Additional blood tests to assess the patient's overall health and nutritional status may include serum electrolytes, a complete blood count, and nutritional markers such as albumin or pre-albumin.

A thorough history and physical examination is essential to assess the patient's suitability for operative therapy with special emphasis on the history of previous surgery since most of these operations are now accomplished laparoscopically. A discussion with the patient regarding the benefits, risks, and long-term side effects of the operation is imperative. This conversation should include discussion of possible conversion to an open procedure, inadvertent or missed injuries, and postgastrectomy syndromes requiring lifestyle and dietary modifications.



SURGERY

In our practice the majority of the elective gastric resections are performed by a laparoscopic approach. Therefore, the laparoscopic approach will be presented in detail, but ultimately, the same operative steps are generally true for an open approach. In the unusual instance when the laparoscopic approach to this procedure is employed in patients presenting with bleeding or perforation, a low threshold for conversion to an open procedure should exist since visualization of important structures will be impaired in these cases.

Positioning

The patient is placed in the supine position and secured to the operative table on which a foot-board is applied to prevent patient slippage as a substantial portion of the procedure is performed in steep reverse Trendelenburg. The patient's arms are usually abducted though they may be tucked at the patient's side depending on the surgeon's and the anesthesiologist's preference. Typical trocar placement for upper gastrointestinal surgery is used as depicted in Figure 7.1. When an open approach is used, an upper midline incision from the xiphoid process to the umbilicus is often sufficient. Alternatively, a bilateral subcostal incision can be used for excellent exposure to the upper abdomen. However, if the patient has had previous abdominal surgery, adhesiolysis in the lower abdomen may prove difficult with the bilateral subcostal incision. Therefore, we prefer a midline incision.

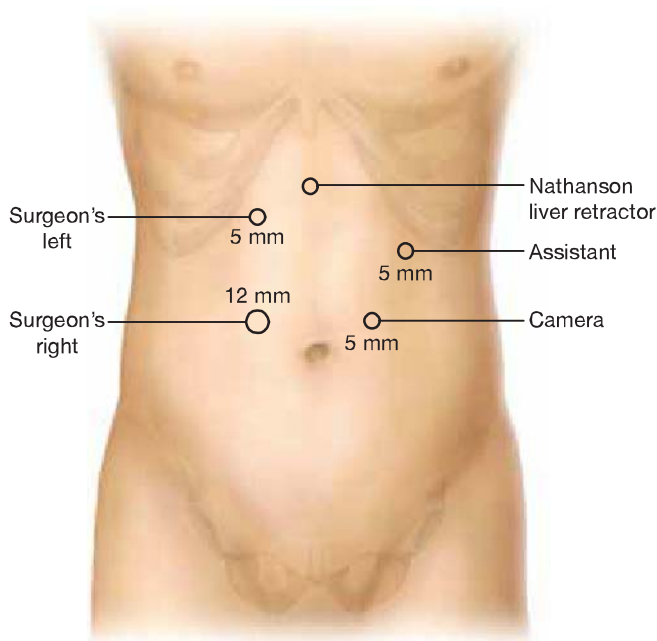


Figure 7.1 Port placement for upper gastrointestinal surgery.

Operative Technique

Following proper positioning, establishment of pneumoperitoneum, and port placement the operative approach is as follows:

- Dissection begins at the esophageal hiatus. The gastrohepatic ligament is incised until the right crus of the diaphragm is well visualized from apex to insertion (Fig. 7.2).
- The phrenoesophageal ligament is divided from right to left anterior to the esophagus, until the left crus is visualized.
- Circumferential dissection of the esophagus is performed to free approximately 4 cm of esophagus. The right and left vagus nerves are identified along the esophagus. The right vagus nerve is typically a relatively large cord-like structure and readily identified in the periesophageal tissue directly adjacent to the posterior surface of the esophagus. The more diminutive left vagus nerve is often directly applied on the anterior surface of the esophagus and less easily identified (Fig. 7.3).
- Each nerve is then clipped and divided using a 5 mm clip applier and laparoscopic scissor. Approximately 1 cm of the nerve is removed and sent to pathology for histologic confirmation. To ensure complete vagotomy, 4 cm of the esophagus should be skeletonized of nerve-like tissue with particular attention to a criminal nerve of Grassi, which is the first branch of the right vagus nerve.

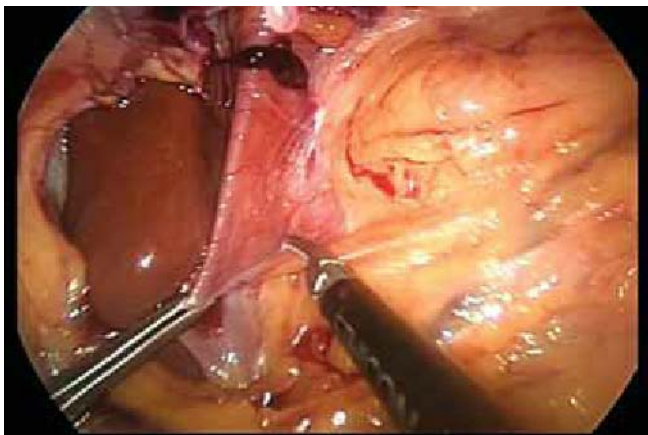


Figure 7.2 Dissection of the right crus of the diaphragm.

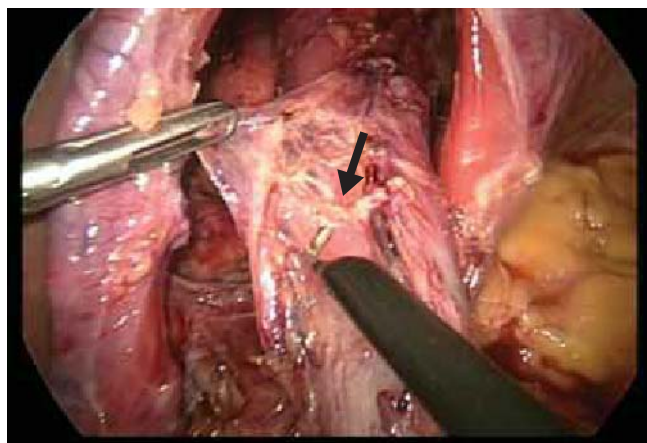


Figure 7.3 The anterior vagus nerve is mobilized from the esophagus (arrow). The nerve is then clipped and divided using a clip applicator and laparoscopic scissors.

- To perform the antrectomy the lesser sac is entered by dividing the gastrocolic omentum with a laparoscopic energy device. Once in the lesser sac, the omentum and the retrogastric attachments are divided toward the pylorus until the origin of the right gastroepiploic vessel is seen (Fig. 7.4). Circumferential dissection around this vascular pedicle is performed, and the pedicle is divided with a laparoscopic 60 mm linear stapler with a vascular load (2.5 mm staples) (Fig. 7.5).
- Division of the gastrocolic omentum is continued orally until the selected point of gastric transection (Fig. 7.6). For antrectomy, approximately one-third of the stomach is removed.
- Along the lesser curvature of the stomach, the right gastric artery is identified and divided using a clip applicator and laparoscopic scissors or a stapling device as desired. Small arteries may also be divided with a laparoscopic energy device (Fig. 7.7).
- A tunnel posterior to the first portion of the duodenum is created. In the presence of chronic inflammation this can be difficult and time consuming (Fig. 7.8). However, careful identification of surrounding structures will guide a safe dissection. Certainly, the location of the common hepatic artery and common bile duct should be known prior to division of the duodenum.
- The duodenum can be divided just distal to the pylorus with a laparoscopic 60 mm linear stapler with 3.5 mm staples (blue load) (Fig. 7.9). The staple line should be examined carefully for defects; if integrity of the staple line is in question or significant inflammation or thickening is present, the staple line should be reinforced with suture.
- The stomach is divided proximally to complete antrectomy, using a linear 60 mm stapler with 3.5 mm staples (Fig. 7.10).



Figure 7.4 After dividing the gastrocolic omentum and entering the lesser sac, the origin of the right gastroepiploic vessels can be found near the pylorus.

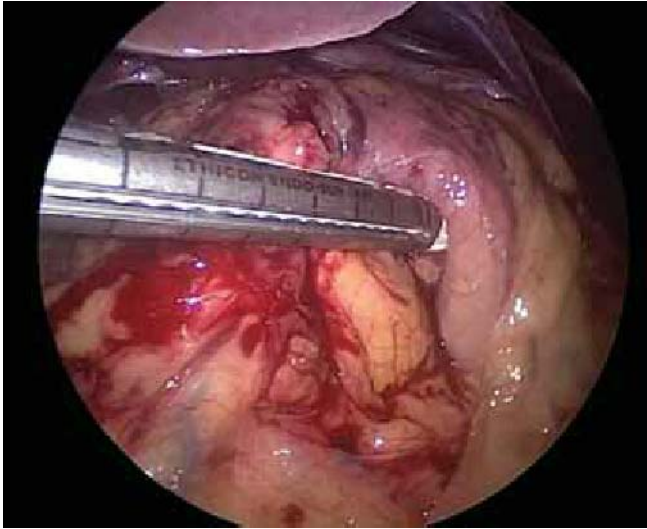


Figure 7.5 Division of the right gastroepiploic pedicle is performed with a laparoscopic linear stapler.

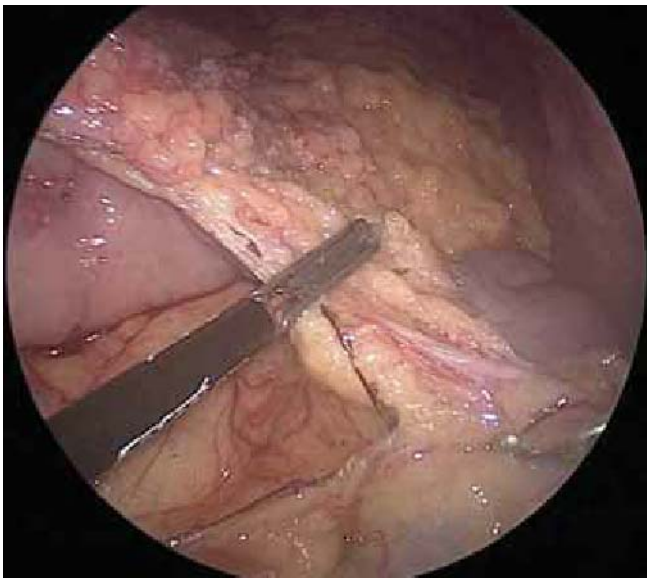


Figure 7.6 Cephalad division of the gastrocolic omentum using a laparoscopic energy device. Transections are continued proximally until the selected point of gastric transection.

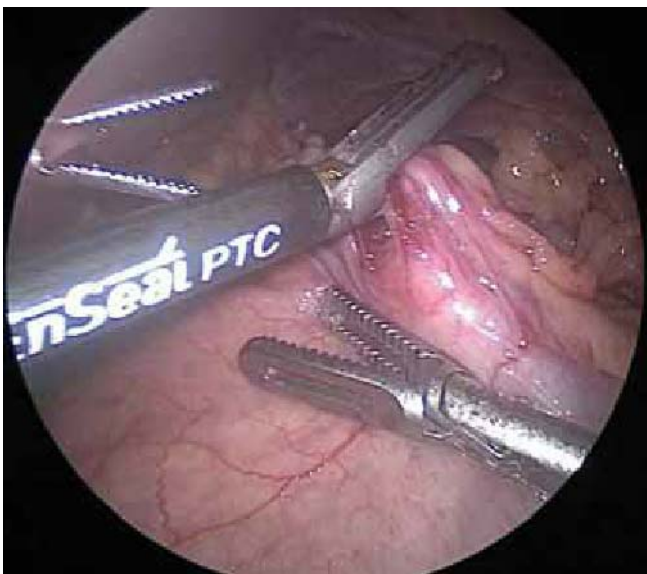


Figure 7.7 Division of the right gastric artery using a laparoscopic energy device.



Figure 7.8 A tunnel posterior to the first portion of the duodenum is created to accommodate the laparoscopic stapler.

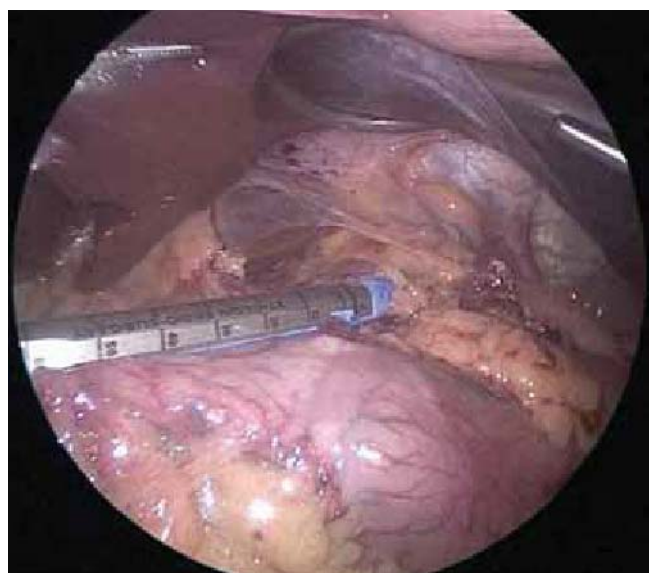


Figure 7.9 The duodenum is divided just distal to the pylorus with a laparoscopic 60 mm linear stapler with 3.5 mm staples (blue load).

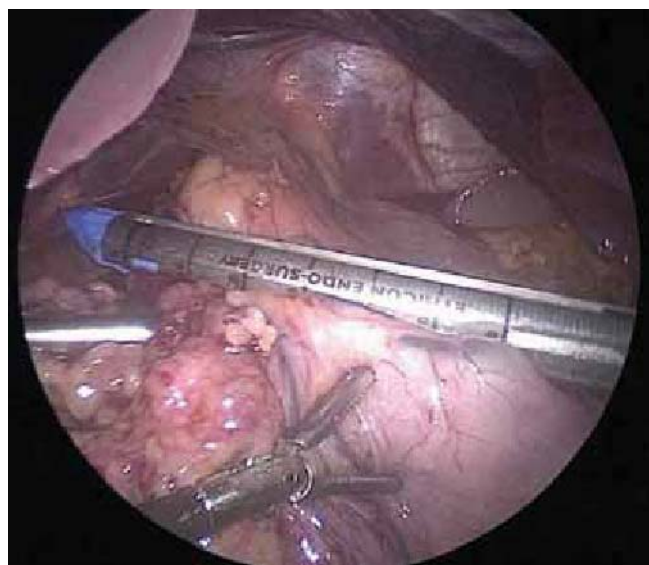


Figure 7.10 The proximal stomach is divided to complete the antrectomy. Approximately one-third of the stomach is removed.



Figure 7.11 View of the ligament of Treitz.

- The specimen is placed in a plastic bag in the right upper quadrant for later retrieval.
- The Roux limb is then created. The operating table is returned to a neutral position. The transverse colon is elevated and the ligament of Treitz is identified (Fig. 7.11). The bowel is run distally for approximately 30 cm where the jejunum is divided using a linear stapler with 2.5 or 3.5 mm staples (white or blue load depending on bowel thickness) (Fig. 7.12).
- The mesentery is divided using an energy device down to its base. Alternatively, a linear stapler with 2 mm staples (gray load) can be used to divide the mesentery (Fig. 7.13).
- A 40 to 50 cm Roux limb is then created and a side-to-side jejunojejunostomy is created using a linear stapler for the posterior and anterior rows, respectively (Fig. 7.14).
- The mesenteric defect is closed using a running 2-0 silk to prevent formation of internal hernia (Fig. 7.15).



Figure 7.12 The jejunum is divided approximately 30 cm distally to the ligament of Treitz. Prior to division, the biliopancreatic limb was marked with an absorbable suture to provide limb identification and orientation.



Figure 7.13 The small bowel mesentery is divided down to its base using a laparoscopic stapler with 2 mm staples. Alternatively, a laparoscopic energy device can be used to divide the mesentery safely.



Figure 7.14 After creating a 40 to 50 cm Roux limb, a side-to-side jejunojejunostomy is created using a linear stapler for the posterior and anterior rows, respectively.



Figure 7.15 The mesenteric defect is closed using a running 2-0 silk to prevent formation of internal hernia.